
Reshaping Reasoning in LLMs: A Theoretical Analysis of RL Training Dynamics through Pattern Selection

Anonymous Author(s)

Affiliation

Address

email

Abstract

1 Reinforcement learning (RL) has become a broadly effective paradigm for strength-
2 ening the reasoning capabilities of modern language models, yet the training
3 dynamics that underlie these gains remain poorly understood across the many
4 settings in which RL is deployed. In this work, we develop a unifying explanation
5 of the RL training process that combines systematic empirical analysis with rigorous
6 theoretical modeling. First, through reasoning-pattern-level and token-level
7 analysis that spans models of different scales and a range of reasoning tasks and
8 mathematical domains, we show that, while individual reasoning patterns exhibit
9 largely stable success rates throughout training, RL predominantly optimizes a
10 sparse subset of critical tokens, thereby reshaping the distribution over reasoning
11 patterns and, in turn, model performance. Building on these empirical insights, we
12 introduce a general theoretical framework that captures the training dynamics of
13 RL under two representative and widely used reward families: verifiable rewards
14 (RLVR) and the model’s own internal feedback (RLIF). For RLVR, we characterize
15 the training dynamics through two representative regimes: one in which models
16 readily converge to optimal reasoning strategies, and another in which optimization
17 becomes challenging, revealing that the reasoning quality of the base model is a key
18 determinant of convergence behavior across the spectrum of base models encountered
19 in practice. For RLIF, we explain how internal rewards can initially improve
20 performance yet may ultimately drive degradation as training continues. Extensive
21 experiments across diverse models, tasks, and RL approaches validate our findings,
22 advancing both the theoretical understanding and the practical application of RL
23 for language model enhancement.

24 1 Introduction

25 Recently, state-of-the-art reasoning models such as Gemini2.5 [8], Qwen3 [40], and DeepSeek-R1
26 [14] have demonstrated exceptional performance across a broad range of complex logical tasks,
27 including mathematics [30, 44, 41] and programming [49, 40, 8]. Reinforcement learning (RL) has
28 emerged as a key technique behind this success, repeatedly showing the potential to elevate the
29 capabilities of diverse model families to a new level.

30 The broad success of RL has triggered a growing body of research into its underlying mechanisms
31 for LLMs. By comparing pass@k performance, Yue et al. [42] show that models post-trained with
32 RL struggle to surpass base models, suggesting that RL may not elicit fundamentally new reasoning
33 patterns. From an entropy perspective, Cui et al. [9], Zhang et al. [45] theoretically prove that
34 RL-based methods can reduce policy entropy, with Cui et al. [9] also empirically establishing a
35 connection between model performance and policy entropy. Wang et al. [33], Huan et al. [17] further
36 demonstrate that RL primarily optimizes a sparse subset of critical tokens. Despite these varied and
37 complementary efforts to interpret RL mechanisms, the training dynamics that govern the RL process

38 remain incompletely understood, both empirically and theoretically, across the range of models and
39 reward schemes used in practice.

40 To understand the RL training process, we first present a systematic reasoning-pattern-level and
41 token-level analysis that spans RL training across models of varying scale and a diverse collection of
42 reasoning tasks and mathematical domains. Specifically, focusing on the training dynamics, we not
43 only examine the ranking shifts across training, but also use both LLM-based and rule-based methods
44 to extract and classify reasoning patterns from models’ responses, analyzing the corresponding
45 success rates and distributions throughout training. Compared with previous works [17, 42], our
46 experiments provide clearer and more broadly compelling evidence that RL primarily optimizes a
47 sparse subset of critical tokens, thereby *reshaping reasoning pattern distributions* to affect model
48 performance. Moreover, we consistently find that the intrinsic success rate of individual patterns
49 remains relatively stable across these settings. These experimental insights motivate us to develop a
50 general mathematical framework for theoretically understanding the RL training process.

51 Building on these empirical findings, we further develop a theoretical framework that conceptualizes
52 reasoning quite generally as a two-stage question-reason-answer process $q \rightarrow r \rightarrow a$: (1) reasoning
53 pattern selection based on the question, i.e., $\pi(r|q)$, and (2) answer generation based on the chosen
54 pattern, i.e., $\pi(a|r, q)$. Because this abstraction does not depend on any particular task or model, it
55 applies uniformly across the reasoning settings we study. Using this framework, we theoretically
56 analyze the training dynamics of RL in LLMs under two representative reward types that together
57 cover much of the current RL landscape: verifiable rewards (RLVR) [14, 30] and the model’s internal
58 feedback (RLIF) [47, 4]. For RLVR, we show that it can converge to the reasoning pattern r^* with the
59 highest success rate, which precisely matches our empirical observations. Moreover, we characterize
60 two distinct convergence regimes that together span the range of base models seen in practice: models
61 with strong initial reasoning quality exhibit rapid convergence to optimal patterns, while weaker
62 models face entanglement-stage optimization challenges. For RLIF, we explain why RL with internal
63 rewards can improve model performance, and we also show that RLIF may ultimately converge to a
64 state with worse performance than the base model, aligning with our empirical findings. Additional
65 experiments across models and tasks validate our theoretical analysis.

66 The main contributions of this paper are highlighted as follows:

- 67 • We conduct systematic reasoning-pattern-level and token-level analysis across the RL training
68 process, spanning multiple model scales, reasoning tasks, and mathematical domains. Through ex-
69 amining next-token-prediction ranking shifts, combined with LLM-based and rule-based reasoning
70 pattern analysis, we provide clearer and more broadly applicable evidence for understanding RL
71 training dynamics than previous works [17, 42]. Our experiments demonstrate that RL primarily
72 optimizes a sparse subset of critical tokens, thereby *reshaping reasoning pattern distributions* to
73 affect model performance, while the intrinsic success rate of individual patterns remains relatively
74 stable during training.
- 75 • We develop a general two-stage mathematical framework that models reasoning as $q \rightarrow r \rightarrow a$
76 (question-reason-answer) and use it to theoretically analyze the training dynamics of two repre-
77 sentative families of RL-based approaches. For RLVR, we prove convergence to the reasoning
78 pattern with the highest success rate and characterize two distinct regimes that jointly cover strong
79 and weak base models: rapid convergence for strong base models versus optimization challenges
80 during entanglement stages for weaker ones. For RLIF, we provide theoretical justification for the
81 performance improvements at early training stages, and explain why such methods may ultimately
82 converge to worse performance than the base model.
- 83 • We validate our theoretical analysis through additional case studies and numerical simulations.
84 More broadly, our framework offers practical insights for understanding and improving RL-based
85 LLM post-training across models and tasks, bridging the gap between empirical observations and a
86 principled understanding of RL training dynamics in LLMs.

87 2 Related Works

88 **Reinforcement Learning for LLMs** Reinforcement learning has demonstrated remarkable success
89 across a wide array of applications for large language models (LLMs), ranging from aligning models
90 with human preferences [25, 48] to solving complex mathematical and programming tasks [30, 20, 22].

A central component of RL is the reward model, which traditionally relies on human-annotated datasets [26, 25, 2] together with extensive training. More recently, the paradigm has broadened toward leveraging rewards that are easier to obtain and to verify, as in reinforcement learning with verifiable rewards (RLVR) [30, 40, 14] and reinforcement learning with internal feedback (RLIF) [47, 4]. This success has, in turn, spurred the development of a growing family of RL algorithms, including GRPO [30], DAPO [41], and VAPO [43].

Theoretical Analyses and Mechanism Interpretation of RL for LLMs. The broad success of RL in LLMs has triggered extensive research into theoretical analyses and mechanism interpretation. Initial efforts mainly focused on high-level analysis of RL, such as how to design better rewards [28, 18, 38] and how to develop novel training algorithms [37, 11, 21]. As RL paradigms have broadened toward simpler reward structures like RLVR and RLIF, researchers have sought to understand the fundamental reasons behind their effectiveness from a variety of perspectives. Shao et al. [29] revealed that spurious rewards enhance reasoning by unlocking latent skills from pretraining, while Agarwal et al. [4] explained performance improvements through entropy minimization. Gandhi et al. [13] identified specific cognitive behaviors, such as backtracking, that contribute to improved reasoning capabilities. Further insights came from [46], who demonstrated that RL fine-tuning amplifies pre-trained behaviors, leading to convergence towards dominant output formats, a phenomenon further analyzed by Wang et al. [34], Cui et al. [10] through the lens of entropy collapse. Complementing these lines of work, our study analyzes RL training dynamics from the unifying perspective of reasoning pattern selection, supported by a general mathematical framework with theoretical guarantees that apply across reward types.

3 Preliminaries

Reinforcement Learning for LLMs We adopt a general formulation that applies to a broad class of language models and reward schemes. Let π_θ be a language model with parameters θ , which serves as the policy to be optimized. Given an input question $\mathbf{x} = (x_0, x_1, \dots, x_n)$, the policy π_θ generates an answer $\mathbf{y} = (y_0, y_1, \dots, y_m)$. The optimization objective can be formulated as:

$$\phi_{\text{RL}}(\theta) = \mathbb{E}_{\mathbf{x} \sim \mathcal{D}, \mathbf{y} \sim \pi_\theta(\cdot | \mathbf{x})} [r_\phi(\mathbf{x}, \mathbf{y})] - \beta \mathbb{D}_{\text{KL}} [\pi_\theta(\mathbf{y} | \mathbf{x}) \parallel \pi_{\text{ref}}(\mathbf{y} | \mathbf{x})], \quad (3.1)$$

where π_{ref} is the base reference policy, and β is a hyperparameter that controls the KL divergence to prevent excessive deviation between π_θ and π_{ref} .

The reward function $r_\phi(\mathbf{x}, \mathbf{y})$ can be instantiated in many forms, such as a trained reward model [25] or a rule-based scoring function [30]. To keep our analysis both concrete and broadly representative of current practice, we focus on two reward families that together span much of the modern RL landscape: verifiable rewards (RLVR) [14, 30] and the model’s internal feedback (RLIF) [47, 4].

In RLVR, the reward function $r_\phi(\mathbf{x}, \mathbf{y})$ directly evaluates whether the answer $\mathbf{y}$ matches the correct answer to question $\mathbf{x}$, a criterion that is applicable whenever answers can be automatically verified. A typical reward function is defined as:

$$r_\phi(\mathbf{x}, \mathbf{y}) = \begin{cases} 1 & \text{if } \mathbf{y} = \text{the ground truth of } \mathbf{x}, \\ 0 & \text{otherwise.} \end{cases} \quad (3.2)$$

Complementing RLVR, RLIF [47, 4] leverages rewards derived solely from intrinsic, model-internal signals, and is thus applicable even in settings where external verification is unavailable. Following Zhao et al. [47], we consider the RLIF reward as the negative average KL divergence between a uniform distribution U over the vocabulary $\mathcal{V}$ and the model’s next-token distribution

$$r_\phi(\mathbf{x}, \mathbf{y}) = \frac{1}{|\mathbf{y}|} \sum_{i=1}^{|\mathbf{y}|} \mathbb{D}_{\text{KL}}(U \parallel \pi_\theta(\cdot | \mathbf{x}, \mathbf{y}_{<i})) = -\frac{1}{|\mathbf{y}| \cdot |\mathcal{V}|} \sum_{i=1}^{|\mathbf{y}|} \sum_{j=1}^{|\mathcal{V}|} \log(|\mathcal{V}| \cdot \pi_\theta(\cdot | \mathbf{x}, \mathbf{y}_{<i})). \quad (3.3)$$

In this paper, we aim to understand the training dynamics of both RLIF and RLVR through complementary empirical and theoretical analysis, treating these two reward families as representative instances of the broader RL paradigm.

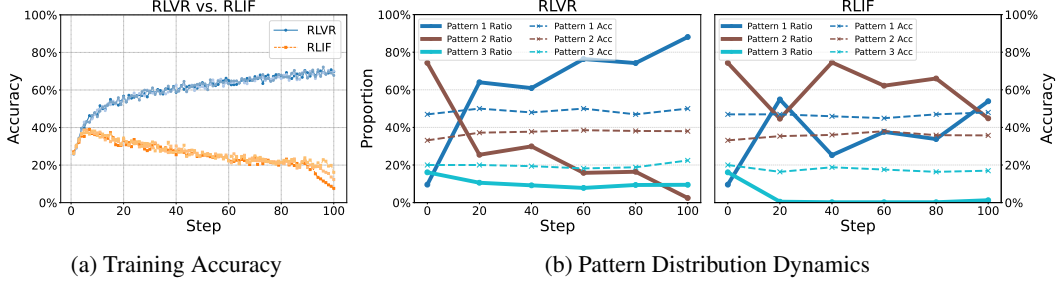


Figure 1: *(a)* The training procedures using RLVR and RLIF, illustrated here on the MATH dataset as a representative case. A key finding, which we observe consistently across settings, is that RLVR provides stable gains, while RLIF causes model performance to initially increase and then decrease. We conduct multiple rounds of experiments for each training paradigm by varying the random seed. *(b)* The reasoning-pattern level comparison. The solid line represents the proportion of a given pattern among all patterns in the responses (left vertical axis), while the dotted line represents the accuracy corresponding to that pattern (right vertical axis). During RLVR, the model gradually adopts patterns with higher accuracy and reduces the use of patterns with lower accuracy, whereas RLIF exhibits unstable training dynamics. For ease of observation, we sort and name the different patterns from high to low according to their corresponding accuracy.

133 4 Experimental Exploration for RL Training

134 To understand the RL training process, we conduct systematic experiments and analysis across
 135 RL training with different reward types, spanning models of varying scale and a diverse set of
 136 reasoning tasks and mathematical domains. Our experiments begin with a high-level overview of the
 137 training procedure, revealing that RLVR yields steady improvements, whereas RLIF can be unstable
 138 and even degrade performance. To dissect these dynamics, we perform a reasoning-pattern level
 139 analysis, which shows that RLVR’s success stems from its tendency to adopt reasoning patterns with
 140 higher success rates, while RLIF fails to specialize in these patterns. Finally, a token-level analysis
 141 investigates the underlying mechanism behind these reasoning pattern dynamics, revealing that they
 142 are driven by changes in the probability ranks of a surprisingly small fraction of tokens. Compared
 143 with previous works [17, 42], our experiments provide clearer and more broadly compelling evidence
 144 of these RL training dynamics, consistently demonstrating across settings that RL primarily optimizes
 145 a sparse subset of critical tokens, thereby *reshaping reasoning pattern distributions* to affect model
 146 performance.

147 4.1 Training Procedure Overview for RLVR and RLIF

148 We first analyze the complete training procedures for both RLVR and RLIF. For a controlled compari-
 149 son, we select Qwen2.5-3B [39] as a representative base model and train it on the MATH dataset
 150 [16], keeping all other settings identical for both algorithms.

151 As illustrated in Figure 1a, the two reward families exhibit markedly different performance trajectories.
 152 RLVR leads to a stable training process, in which model performance continuously improves and
 153 gradually converges throughout training. In contrast, while RLIF initially improves performance
 154 in the early stages, continued training can lead to a performance drop, sometimes yielding a model
 155 worse than the original base model. This qualitative contrast, which we later show to hold broadly
 156 across tasks and models, motivates us to conduct a more systematic, fine-grained empirical analysis
 157 of the underlying RL training dynamics.

158 4.2 Reasoning-pattern Level Analysis

159 To understand the performance disparities observed between RLVR and RLIF, we conduct a fine-
 160 grained analysis at the reasoning-pattern level. Our pattern-extraction procedure is task-agnostic and
 161 therefore applies uniformly across the diverse reasoning tasks we study: we first collect responses
 162 from the base model and employ GPT-4o [19] to group them into distinct categories based on
 163 keywords and logical structure, also generating a description for each pattern. We then use these

classifications to analyze the reasoning pattern distributions and their corresponding success rates throughout the RL training procedure.

Specifically, we analyze the dynamics of reasoning patterns and their accuracy during both RLVR and RLIF training. Figure 1b illustrates our analysis on responses from Number Theory tasks in the MATH dataset [16] as a representative example, with consistent results across a broader range of models and datasets provided in Appendix A. Our analysis reveals three key findings that hold robustly across these settings:

- RLVR-trained models consistently shift towards adopting reasoning patterns with higher success rates, explaining why the model’s overall accuracy steadily improves with RLVR.
- The reasoning pattern distribution for RLIF exhibits unstable training dynamics and fails to specialize in more effective patterns.
- The success rate of any individual reasoning pattern remains stable throughout training for both reward families.

This analysis indicates that the performance difference between RLVR and RLIF stems from their distinct reasoning pattern dynamics rather than from task-specific idiosyncrasies. We therefore conduct further experiments to uncover the underlying mechanisms driving these distributional changes during RL training.

4.3 Token-level Analysis

To understand the mechanisms behind the reasoning pattern dynamics observed during RL training, and inspired by Huan et al. [17], Yue et al. [42], we further conduct a token-level analysis spanning datasets of markedly different difficulty. We first sample responses from the base model for questions drawn from the GSM8K [6], MATH [16], and AIME [7] datasets. Then, for these identical responses, we examine the corresponding token ranks in both the base and the RL-enhanced models. Here, a token’s rank refers to the position of its probability among all vocabulary tokens, given the preceding context. Our analysis, summarized in Table 1, reveals that the ranks of individual tokens remain largely stable regardless of dataset or reward family, with less than 10% of tokens experiencing a rank shift in most cases. This finding indicates that RL selectively modifies the probabilities of a sparse set of critical tokens while leaving the vast majority of the reasoning process unchanged.

Taken together, our experiments provide clean and broadly consistent evidence for RL training dynamics: *RL primarily optimizes a sparse subset of critical tokens, thereby reshaping reasoning pattern distributions to affect model performance. Moreover, the intrinsic success rate of individual patterns remains relatively stable during training.*

These empirical findings, observed consistently across models, tasks, and reward types, not only deepen our understanding of RL training dynamics but also motivate and inform the general mathematical framework we develop for further theoretical analysis.

5 Theoretical Characterization of RL Training Dynamics

Beyond empirical observations, we take a further step toward a theoretical understanding of the RL training process that is not tied to any single task or model. In this section, we first build a general theoretical framework grounded in our experimental insights. We abstract the model’s reasoning procedure as a two-step process: first selecting a reasoning pattern, then performing answer deduction, where RL specializes in optimizing the first step. Because this abstraction is task- and architecture-agnostic, the resulting analysis applies broadly across the reasoning settings we consider. Within this framework, we analyze RL under two representative reward families. For RLVR, we characterize its training dynamics through two regimes that together capture the range of distinct optimization

Table 1: The token ranking change ratio after RL, reported across representative datasets and both reward families.

Task	Method	Step20	Step40	Step60	Step80	Step100
GSM8K	RLVR	5.2%	6.0%	6.6%	7.1%	7.3%
	RLIF	6.8%	8.0%	8.9%	9.5%	10.1%
MATH	RLVR	5.3%	5.8%	6.1%	6.5%	6.6%
	RLIF	6.1%	7.0%	7.7%	8.3%	8.8%
AIME24	RLVR	5.1%	5.7%	5.9%	6.1%	6.3%
	RLIF	5.6%	6.5%	7.4%	8.2%	8.6%

behaviors seen in practice. For RLIF, we explain why RL with internal rewards can improve model performance while also clarifying why such methods may ultimately yield models that perform worse than the base model.

5.1 A Theoretical Framework for Reasoning Models

Abstract Reasoning Process. We formalize the reasoning process as follows: given a question q , the model (1) selects a reasoning pattern r from candidate patterns $\mathcal{R} = \{r_1, r_2, \dots\}$ and (2) generates a final answer $a \in \mathcal{A} = \{a_1, a_2, \dots\}$ accordingly. The model selects reasoning pattern r_i with probability $p(r_i|q)$, and each reasoning pattern has a distinct success rate. Within this framework, we can reformulate the RLIF optimization objective (Eq. 3.3) as follows¹:

$$r_\phi(\{r, a\}, q) := - \left(\frac{1}{|\mathcal{R}|} \sum_{r_i \in \mathcal{R}} \log(|\mathcal{R}| \cdot \pi_\theta(r_i|q)) + \frac{1}{|\mathcal{A}|} \sum_{a_j \in \mathcal{A}} \log(|\mathcal{A}| \cdot \pi_\theta(a_j|q, r)) \right) \quad (5.1)$$

Policy Parameterization. Let $\mathcal{V} = \{q\} \cup \mathcal{R} \cup \mathcal{A}$ denote the vocabulary set and $\mathcal{Y} = (r, a)_{r \in \mathcal{R}, a \in \mathcal{A}}$ represent the set of output sequences. Given a question q , the language model produces a distribution over output sequences $y \in \mathcal{Y}$ autoregressively:

$$(\text{general policy}) \quad \pi_\theta(y|x) = \prod_{l=1}^{|y|} \pi_\theta(y_l|x, y_{<l}) = \prod_{l=1}^{|y|} \text{softmax}(f_\theta(x, y_{<l}))_{y_l}, \quad (5.2)$$

where $f_\theta : \mathcal{V} \rightarrow \mathbb{R}^{|\mathcal{V}|}$ is a function parameterized by θ , and the model predicts the l -th token based on the previous context $y_{<l}$. The next token follows the distribution $\text{softmax}(z)_v := \exp(z_v) / \sum_{v' \in \mathcal{V}} \exp(z_{v'})$ for $z = f_\theta(x, y_{<l})$.

Due to the complexity of practical models, establishing optimization guarantees for understanding RL training dynamics has proven very challenging [3, 23]. Following a well-established line of prior work [27, 24, 10, 45], we adopt a *tabular policy* parameterization, a standard analytical abstraction that can be viewed as a special case of Eq. 5.2 in which each output is assigned its own trainable logit for the corresponding last token, i.e., for $\theta \in \mathbb{R}^{|\mathcal{V}| \times |\mathcal{V}|}$:

$$(\text{tabular policy}) \quad \pi_\theta(y_l|x, y_{<l}) = \pi_\theta(y_l|y_{l-1}) = \text{softmax}(\theta_{:,y_{l-1}})_{y_l}, \quad (5.3)$$

where $\theta_{:,y_{l-1}} \in \mathbb{R}^{|\mathcal{V}|}$ is the column of θ corresponding to y_{l-1} .

Optimization for RL. For our analysis of training dynamics, we consider the tabular policy from Eq. 5.3 with the optimization objective $\phi_{\text{RL}}(\theta)$ in Eq. 3.1. We analyze the policy gradient in the small learning rate limit using gradient flow:

$$\frac{d}{dt} \theta(t) = \nabla \phi_{\text{RL}}(\theta(t)), \quad t \geq 0 \quad (5.4)$$

where $\theta(t)$ represents the parameters of the policy $\pi_{\theta(t)}$ at training time t , initialized with $\theta(0) = \theta_{\text{ref}}$.

Our experiments consistently revealed that success rates for individual reasoning patterns tend to remain stable during training. We attribute this general phenomenon to a structural property shared by such models: optimizing the mapping from questions to reasoning patterns is substantially easier than optimizing the path from reasoning patterns to final answers. To formalize this broadly observed behavior, we introduce the following assumption:

Assumption 5.1. *The success rate for each reasoning pattern $r_i \in \mathcal{R}$ to provide the correct answer r^* , defined as $\pi_\theta(a^*|q, r_i)$, remains constant during training.*

In the remaining part of this section, we adopt Assumption 5.1 and use $p^*(r) = \pi_\theta(a^*|q, r)$ to denote the fixed success rate for the given pattern r . We first derive the optimal policy for the RL objective (Eq. 3.1) under the general autoregressive policy (Eq. 5.2). We then analyze RLVR and RLIF training dynamics using the tabular policy (Eq. 5.3).

¹For simplicity, we omit the normalizing coefficient $1/|y|$ from Eq. 3.3, which can be treated as a constant $1/2$ in our framework.

5.2 The Optimal Policy for RL

Our empirical results demonstrate that RLVR steadily improves model performance by incentivizing reasoning patterns with higher success rates, whereas RLIF yields only unstable improvement. To explain both behaviors within a single treatment, we first characterize the optimal policy for the RLVR and RLIF objectives in a unified form:

Proposition 5.2. *Suppose we maximize the RL objective (Eq 3.1) using a general autoregressive policy (Eq 5.2) and Assumption 5.1 holds. Then, the optimal policy satisfies:*

$$\pi_{\text{opt}}(\mathbf{r}|\mathbf{q}) = \frac{1}{Z} \exp\left(\frac{1}{\beta} R(\mathbf{r})\right) \pi_{\theta_{\text{ref}}}(\mathbf{r}|\mathbf{q}) \text{ for all } \mathbf{r} \in \mathcal{R}, \quad (5.5)$$

where $Z = \sum_{\mathbf{r} \in \mathcal{R}} \exp\left(\frac{1}{\beta} R(\mathbf{r})\right) \pi_{\theta_{\text{ref}}}(\mathbf{r}|\mathbf{q})$ is the normalizing coefficient, and $R(\mathbf{r})$ denotes the reasoning path reward. Specifically:

- For RLVR, $R(\mathbf{r})$ equals the success rate of reasoning pattern $\mathbf{r}$, i.e., $R_{\text{RLVR}}(\mathbf{r}) = p^*(\mathbf{r})$.
- for RLIF, $R(\mathbf{r})$ equals to the confidence for the reasoning pattern $\mathbf{r}$ for final answer distribution, i.e., $R_{\text{RLIF}}(\mathbf{r}) = -\frac{1}{|\mathcal{A}|} \sum_{\mathbf{a} \in \mathcal{A}} \log(\pi_{\theta}(\mathbf{a}|\mathbf{q}, \mathbf{r}))$.

Proposition 5.2 characterizes the optimal solutions for RLVR and RLIF within a single expression. Assuming π_{opt} resides within the general autoregressive policy parameterized space (which always holds for both LLMs and the tabular policies discussed later), the RLVR-optimized model increases the probability of a reasoning pattern $\mathbf{r}$ whenever its probability of deducing the correct answer $p^*(\mathbf{r})$ is high enough that $\exp(\frac{1}{\beta} p^*(\mathbf{r}))/Z > 1$. In contrast, for RLIF, the reward $R_{\text{RLIF}}(\mathbf{r})$ depends only on the confidence of the final answer distribution and thus fails to distinguish the correct answer $\mathbf{a}^*$ from incorrect alternatives. Consequently, while the optimal policy under RLVR consistently leads to better performance, the optimal policy for RLIF carries no such guarantee of improved accuracy over the base model, an asymmetry that our framework captures uniformly for both reward families.

Moreover, since β is typically small (e.g., 0.001) in practice [26, 5, 44], as $\beta \rightarrow 0$, we have:

$$\pi_{\text{opt}}(\mathbf{r}|\mathbf{q}) = \lim_{\beta \rightarrow 0} \frac{1}{Z} \exp\left(\frac{1}{\beta} R(\mathbf{r})\right) \pi_{\theta_{\text{ref}}}(\mathbf{r}|\mathbf{q}) = \begin{cases} 1 & \mathbf{r} = \arg \max_{\mathbf{r}} R(\mathbf{r}) \\ 0 & \text{otherwise} \end{cases}, \quad (5.6)$$

In this limit, the policy converges to a deterministic strategy that always selects the reasoning pattern $\mathbf{r}$ with highest $R(\mathbf{r})$, regardless of how the initialized reference model chooses reasoning patterns. However, due to the non-convexity of the optimization landscape, the dynamics of how RLVR reliably finds high-reward patterns, and why RLIF may exhibit early performance improvements, remain unclear from the optimum alone. To address this in generality, we next analyze the training dynamics of both reward families under a tabular policy (Eq. 5.3), providing deeper insight into the RL training process.

5.3 Training Dynamic Analysis for RLVR

Our experiments demonstrate that RLVR improvements depend heavily on the capacity of the base model, a dependence that holds across the range of models we consider. Here, we focus on the probability of the *optimal reasoning pattern*, i.e., the pattern $\mathbf{r}^*$ with the highest success rate for reaching the correct answer $\mathbf{a}^*$, $\mathbf{r}^* = \arg \max_{\mathbf{r}} p^*(\mathbf{r})$. Through gradient flow, we reveal two distinct regimes in the training dynamics of RLVR that together span the behaviors observed for strong and weak base models alike. In the first regime, the probability of the optimal reasoning pattern $\pi_{\theta}(\mathbf{r}^*|\mathbf{q})$ steadily increases until convergence to 1. In the second regime, the model initially experiences an entanglement stage, in which a suboptimal reasoning pattern $\mathbf{r}' \neq \mathbf{r}^*$ hinders optimization of the optimal reasoning pattern. After this entanglement stage, the model eventually transitions to the dynamics of the first regime and converges to the optimal reasoning pattern $\mathbf{r}^*$.

Theorem 5.3 (Regime 1: Sufficient Condition for Efficient Convergence). *Consider the RLVR (Eq 3.2) with $\beta = 0$ for optimizing objective Eq 3.1, using a tabular policy (Eq 5.3) with Assumption 5.1 holds. Let $\mathbf{r}^*$ be the optimal reasoning pattern, if the overall accuracy of the initialized model $\pi_{\theta_{\text{ref}}}$, defined as $\text{ACC}_{\theta_{\text{ref}}} = \sum_{\mathbf{r} \in \mathcal{R}} \pi_{\theta_{\text{ref}}}(\mathbf{r}|\mathbf{q}) p^*(\mathbf{r})$, satisfies:*

$$(\text{Regime 1}) \quad \text{ACC}_{\theta_{\text{ref}}} > p^*(\mathbf{r}) \text{ for all } \mathbf{r} \in \mathcal{R}, \mathbf{r} \neq \mathbf{r}^*, \quad (5.7)$$

then for any $\epsilon > 0$, there exists $T_1 = \mathcal{O}(\frac{1}{\epsilon})$ such that for $t > T_1$, we have $1 - \pi_{\theta(t)}(\mathbf{r}^*|\mathbf{q}) < \epsilon$.

Theorem 5.3 covers the regime in which the base reference model is sufficiently strong that its overall accuracy exceeds the success rate of all non-optimal reasoning patterns (Eq 5.7). In this regime, RLVR efficiently guides the model to select the optimal reasoning pattern at rate $\mathcal{O}(1/\epsilon)$, achieving a high overall accuracy (close to the success rate of the optimal reasoning pattern).

However, practical experience across many settings shows that RLVR optimization can be challenging [44, 36], a difficulty that typically arises when the base reference model is less powerful. We therefore consider the second regime, in which the model initially experiences an entanglement stage, and the *suboptimal reasoning pattern with the second-highest success rate*, defined as $\mathbf{r}' = \arg \max_{\mathbf{r}, \mathbf{r} \neq \mathbf{r}^*} p^*(\mathbf{r})$, slows the optimization of the optimal reasoning pattern:

Theorem 5.4 (Regime 2: Slow Convergence for optimal reasoning pattern). *Consider the RLVR (Eq 3.2) with $\beta = 0$ for optimizing objective Eq 3.1, using a tabular policy (Eq 5.3) with Assumption 5.1 holds. Let $\mathbf{r}^*$ and $\mathbf{r}'$ be the optimal and second optimal reasoning patterns, if the overall accuracy of the initialized model satisfies:*

$$(Regime\ 2) \quad p^*(\mathbf{r}') > \text{ACC}_{\theta_{\text{ref}}} > p^*(\mathbf{r}) \text{ for all } \mathbf{r} \in \mathcal{R}/\{\mathbf{r}^*, \mathbf{r}'\}, \quad (5.8)$$

then there exists:

$$T_0 = \frac{1}{2 - 2\pi_{\theta_{\text{ref}}}(\mathbf{r}'|q)} \left((C_1 \cdot \gamma_{\pi_{\text{ref}}})^{2C_2 \cdot \gamma_{\pi_{\text{ref}}}} - 1 \right), \text{ where } \gamma_{\pi_{\text{ref}}} := \sum_{\mathbf{r} \in \mathcal{R}/\{\mathbf{r}'\}} \frac{\pi_{\theta_{\text{ref}}}(\mathbf{r}|q)}{\pi_{\theta_{\text{ref}}}(\mathbf{r}^*|q)} \quad (5.9)$$

with constants C_1, C_2 depending on the success rates of reasoning patterns, such that for we can guarantee the model transform from regime 2 (Eq 5.8) to regime 1 (Eq 5.7), i.e., $\text{ACC}_{\theta(t)} > p^*(\mathbf{r}), \forall \mathbf{r} \in \mathcal{R}, \mathbf{r} \neq \mathbf{r}^*$ for $t \geq T_0$.

In Theorem 5.4, we analyze a representative regime in which only the success rates of the optimal and suboptimal reasoning patterns exceed the overall accuracy. We adopt this setting for analytical clarity, and we expect the conclusions to extend naturally to the more general case in which *at least* two reasoning patterns achieve higher success rates than the average, albeit at the cost of a more involved analysis.

Additionally, the critical insight of Theorem 5.4 is that RLVR may require T_0 time steps before the overall success rate exceeds the success rate of the suboptimal reasoning pattern, i.e., before reaching the regime of Theorem 5.3 in which the selection probability of the optimal reasoning pattern is sufficiently large. While Theorem 5.3 shows that the convergence time T_1 is polynomial in $1/\epsilon$, the time step T_0 in Theorem 5.4 may grow super-exponentially in $\gamma_{\pi_{\text{ref}}}$ —the ratio between the total success rate through $\mathcal{R}/\{\mathbf{r}'\}$ and the success rate through $\mathbf{r}^*$. When the base model assigns a very small selection probability to the optimal reasoning pattern, $\gamma_{\pi_{\text{ref}}}$ can be very large, leading to a prohibitively large T_0 . Consequently, it can take an extremely long training period for the model to select the optimal reasoning pattern with reasonably large probability—a phenomenon we refer to as the entanglement stage, and one that we expect to arise broadly whenever weak base models are trained with RLVR.

We provide additional experiments and case studies across settings to further illustrate these two regimes in Sec 6.

5.4 Theoretical Explanation for RLIF

In Proposition 5.2, we revealed that RLIF, which considers only the confidence of the final answer distribution, fails to distinguish the correct answer $\mathbf{a}^*$ from incorrect alternatives. This suggests that RLIF may eventually degrade performance relative to the base model, aligning well with the empirical observations we report across tasks and models. Yet prior studies have shown that RLIF can also improve model performance without any external reward Agarwal et al. [4], Zhao et al. [47]. To reconcile these seemingly opposing behaviors, we provide a theoretical explanation for why RLIF can improve the performance of a well-trained LLM at the initial training stage.

First, we consider a well-trained LLM satisfying the following assumption:

Assumption 5.5. *For the base model, the correct answer $\mathbf{a}^*$ has the highest probability across all possible answers, i.e., $\arg \max_{\mathbf{a} \in \mathcal{A}} \sum_{\mathbf{r} \in \mathcal{R}} \pi_{\text{base}}(\mathbf{a}|q, \mathbf{r}) \pi_{\text{base}}(\mathbf{r}|q) = \mathbf{a}^*$.*

Since majority voting [35] is a broadly effective method for improving model performance, and well-trained LLMs typically assign a higher probability to the correct answer $\mathbf{a}^*$ than to incorrect

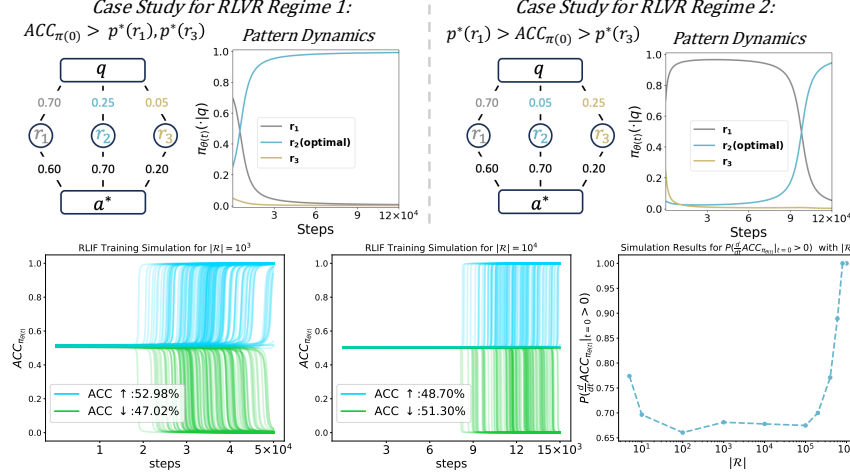


Figure 2: Case studies for RLVR (upper) and training simulations for RLIF (bottom), illustrating the two reward families analyzed in our framework. The upper panels demonstrate the two distinct training regimes predicted for RLVR: rapid convergence (left) and the entanglement phase (right). The bottom panels show the probabilities of model convergence to different states (left) and the probability of initial performance gains (right) for RLIF across varying $|\mathcal{R}|$.

answers, Assumption 5.5 is likely to hold for a wide range of modern LLMs. Under this constraint, we consider the case where $|\mathcal{A}| = 2$, the success rate for each reasoning path follows a uniform distribution, and the reasoning pattern selection is high-entropy². In this representative case, we analyze the overall accuracy dynamics at $t = 0$, yielding the following theorem:

Theorem 5.6 (RLIF Increases Overall Accuracy at Initialization). *Consider the RLIF (Eq 5.1) with $\beta = 0$ for optimizing objective Eq 3.1, using a tabular policy (Eq 5.3) with Assumption 5.1 and 5.5 hold, $\pi_{\theta_{\text{base}}}(\mathbf{r}|q) = \frac{1}{|\mathcal{R}|}$ for all $\mathbf{r} \in \mathcal{R}$, $|\mathcal{A}| = 2$ and success rate for each reasoning path follows a uniform distribution $p^*(\mathbf{r}) \sim U[0, 1]$. Then when $|\mathcal{R}| \rightarrow +\infty$, the following holds:*

1. The accuracy derivative at $t = 0$ is positive with probability 1, i.e. $P(\frac{d}{dt} \text{ACC}_{\theta(t)}|_{t=0} > 0) = 1$.
2. With probability $p = 0.5$, that $\arg \max_{\mathbf{r}} R_{\text{RLIF}}(\mathbf{r}) = \arg \min_{\mathbf{r}} p^*(\mathbf{r})$.

The first result in Theorem 5.6 shows that overall accuracy increases at initialization, offering a theoretical explanation for RLIF’s early performance gains. As training progresses, however, Proposition 5.2 implies that when $\beta = 0$ the policy converges deterministically to the reasoning pattern $\mathbf{r}$ that maximizes $R_{\text{RLIF}}(\mathbf{r})$. The second result in Theorem 5.6 reveals that, with probability 0.5, this maximizing pattern coincides with the one that *minimizes* the success rate: $\arg \max_{\mathbf{r}} R_{\text{RLIF}}(\mathbf{r}) = \arg \min_{\mathbf{r}} p^*(\mathbf{r})$. In other words, there is a 50% chance the model converges to the least accurate reasoning path. Together, these two results paint a coherent and general picture: while RLIF initially improves model performance (the accuracy derivative is positive), continued training may drive the model toward states that favor reasoning paths with very low accuracy, ultimately yielding performance worse than the base model—an explanation that accounts for the unstable RLIF behavior observed across our experiments.

6 Case Studies and Numerical Simulations

To further validate and interpret our theoretical findings, we present case studies and training simulations that illustrate our theorems across both reward families. For RLVR, we demonstrate the two distinct training regimes discussed in Theorems 5.3 and 5.4, as illustrated in Fig. 2 (upper). For RLIF, we conduct multiple training simulations in which each reasoning path’s success rate follows a uniform distribution, validating our analysis of RLIF’s behavior both during the initial training steps and at convergence, as shown in Fig. 2 (bottom).

²Previous work [33] reveals that RL primarily optimizes high-entropy tokens. Here, we consider the highest-entropy case: $\pi_{\theta_{\text{base}}}(\mathbf{r}|q) = 1/|\mathcal{R}|$ for all $\mathbf{r} \in \mathcal{R}$.

6.1 Case Studies for RLVR

In Section 5.3, we identified two distinct regimes that together span RLVR’s training dynamics. In the first regime (Eq 5.7), the probability of selecting the optimal reasoning pattern $\pi_\theta(r^*|q)$ monotonically increases until convergence. In the second regime (Eq 5.8), the model undergoes an initial entanglement phase and requires a substantially longer training period before transitioning to the dynamics of the first regime and converging to the optimal reasoning pattern r^* . We examine both regimes in turn:

- As demonstrated in Fig. 2 (upper left), when the reference model satisfies the conditions in Eq 5.7, so that the overall accuracy exceeds the success rates of all non-optimal reasoning patterns, the model converges rapidly to the optimal reasoning pattern.
- In the second case, corresponding to Theorem 5.4, when a suboptimal reasoning pattern r' achieves a higher success rate than the initial overall accuracy, the model first experiences an entanglement phase. A transition period T_0 (defined in Theorem 5.4) must elapse before the model enters the rapid optimization phase characteristic of the first regime. Fig. 2 (upper right) illustrates a representative scenario in which $\gamma_{\pi_{\text{ref}}}$ (defined in Eq 5.9) is large ($\gamma_{\pi_{\text{ref}}} = 6$, since $(0.25 + 0.05)/0.05 = 6$). In such cases, the transition time T_0 , which scales with $\gamma_{\pi_{\text{ref}}}^{\gamma_{\pi_{\text{ref}}}}$, becomes prohibitively long, substantially delaying convergence to the optimal reasoning pattern. The pattern dynamics in Fig. 2 clearly exhibit this entanglement stage, aligning well with our theoretical prediction.

6.2 Numerical Simulations For RLIF

In Section 5.4, we demonstrated that, under Assumption 5.5 and the base model distribution specified in Theorem 5.6, as $|\mathcal{R}| \rightarrow \infty$ the accuracy derivative at $t = 0$ is positive with probability 1, while there is a 50% probability of convergence to the least accurate reasoning path. Our simulations validate these findings across a range of $|\mathcal{R}|$. Fig. 2 (bottom left) presents RLIF training simulations for varying $|\mathcal{R}|$. The results confirm that the probability of convergence to an improved state ($ACC \uparrow$) approximately equals the probability of convergence to a degraded state ($ACC \downarrow$), validating the second result in Theorem 5.6. Moreover, our examination of initial-step performance, shown in Fig. 2 (bottom right), demonstrates that for large $|\mathcal{R}|$ the probability of initial performance improvement approaches 1, supporting the first result in Theorem 5.6.

7 Conclusions and Limitations

This work analyzes reinforcement learning dynamics in LLMs through complementary empirical investigation and theoretical modeling. Grounded in experiments that span multiple model scales, reasoning tasks, and mathematical domains, we develop a general mathematical framework and instantiate it for two representative reward families (RLVR and RLIF), validating our findings through case studies and simulations. Because the framework is task- and architecture-agnostic, we expect its qualitative conclusions to transfer broadly to other reasoning settings and reward schemes. At the same time, our analysis has limitations that warrant further investigation and point to natural avenues for extending its reach: the interpretability of LLM-identified reasoning patterns merits additional validation, the theoretical framework could be extended to richer real-world reasoning scenarios, and the current analysis could be generalized beyond the specific base-model assumptions we adopt.

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A Additional Experiments

A.1 Pattern Distribution Dynamics on RL-enhanced Qwen2.5-3B

In Section 4.2, we sampled Number Theory tasks from MATH to analyze the dynamics of reasoning patterns. To confirm that these dynamics are not specific to a single domain, we additionally analyze Geometry tasks (Figure 3). During RLVR, the model again gradually adopts the reasoning patterns with higher accuracy, whereas the pattern selection for RLIF remains unstable. This consistency across domains reinforces the empirical and theoretical results presented in the main paper.

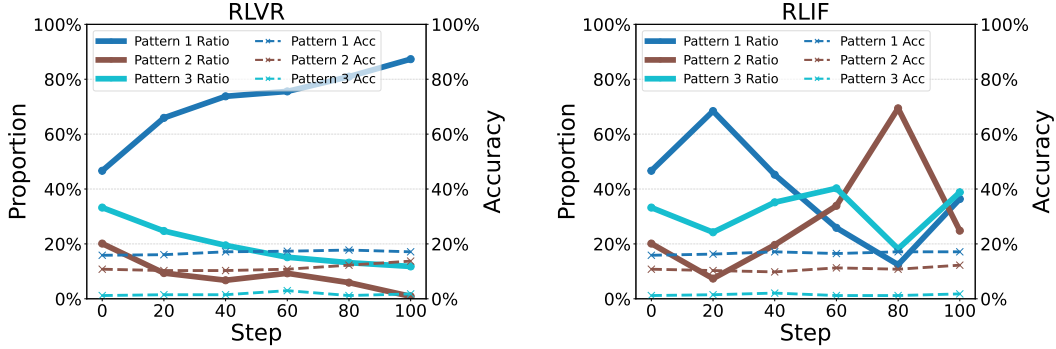


Figure 3: Dynamics of the pattern distribution and corresponding accuracy of RL-enhanced models on Geometry tasks, mirroring the behavior observed on Number Theory. Under RLVR, the model tends to choose the pattern with the highest accuracy (Pattern 1), whereas RLIF remains unstable.

A.2 Qwen-2.5-7B-Instruct vs. Qwen-2.5-7B-SimpleRL-Zoo

To assess how broadly our findings apply across different models, we conduct supplementary experiments that further validate our experimental insights and theoretical conclusions (Figure 4). For these experiments, we use Qwen-2.5-7B-Instruct [39] as the base model and compare it with Qwen-2.5-7B-SimpleRL-Zoo [44], an open-source variant enhanced with RLVR for mathematical reasoning. Our evaluation spans a wide range of mathematical domains—number theory, geometry, algebra, calculus, and counting and probability—using challenging problems sampled from the MATH dataset [16], and we further extend the analysis to complex problems from AMC23 [1]. Across all of these domains, the results consistently support our earlier findings: RLVR enhancement increases the frequency of high-accuracy reasoning patterns while making less effective patterns appear less often. Detailed task-specific analyses are provided in Appendix B.3.

Across this diverse collection of tasks, reasoning pattern analysis reveals that patterns with the highest success rates consistently become more prevalent after RLVR enhancement, reinforcing our findings from Section 4 and Section 5. Notably, for tasks such as Algebra and AMC23-19 in Figure 4, the success rates of individual reasoning patterns remain stable, which not only aligns with our earlier observations but also provides broad empirical support for Assumption 5.1 in our theoretical analysis.

A.3 Qwen-2.5-32B-Instruct vs. QwQ-32B

To assess whether our theory extends to substantially larger models, we conduct experiments on QwQ-32B [32]. This model is based on Qwen-2.5-32B-Instruct [39] and greatly enhances reasoning ability through RLVR. We examine the shift in reasoning patterns between these two models on four tasks that span several domains and benchmarks: number theory and geometry tasks from the MATH dataset [16], a task from AIME 2024 [7], and a task from OlympiadBench [15].

We observe that most of QwQ-32B’s reasoning patterns correspond to the most accurate patterns of Qwen-2.5-32B, aligning well with our theoretical results and indicating that the pattern-selection mechanism persists at this larger scale. We do not report the accuracy of QwQ because we rely on the models’ APIs for testing and cannot extract the CoT data needed to let the base model continue generating answers; without it, the accuracy of the reasoning patterns (as opposed to the accuracy of the model itself) cannot be obtained. For details, see the “Accuracy Analysis” in Appendix B.2.

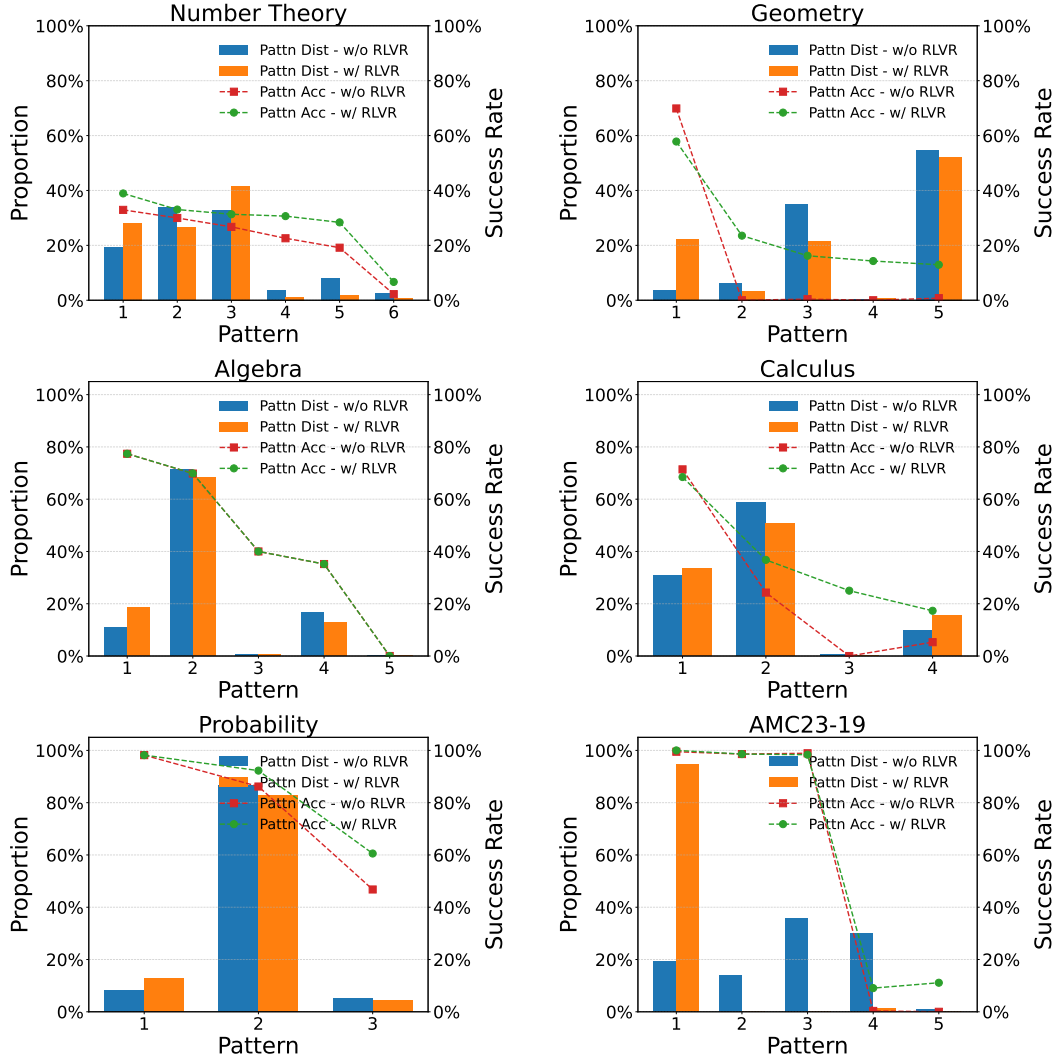


Figure 4: Evaluation results for reasoning patterns and their corresponding success rates for models with/without RLVR enhancement across a broad set of additional tasks, all of which align well with our experimental insights and theoretical conclusions. The bars (Pattern Dist) represent the proportion of each pattern among all patterns, and the dots (Pattern Acc) represent the accuracy or success rate corresponding to that pattern.

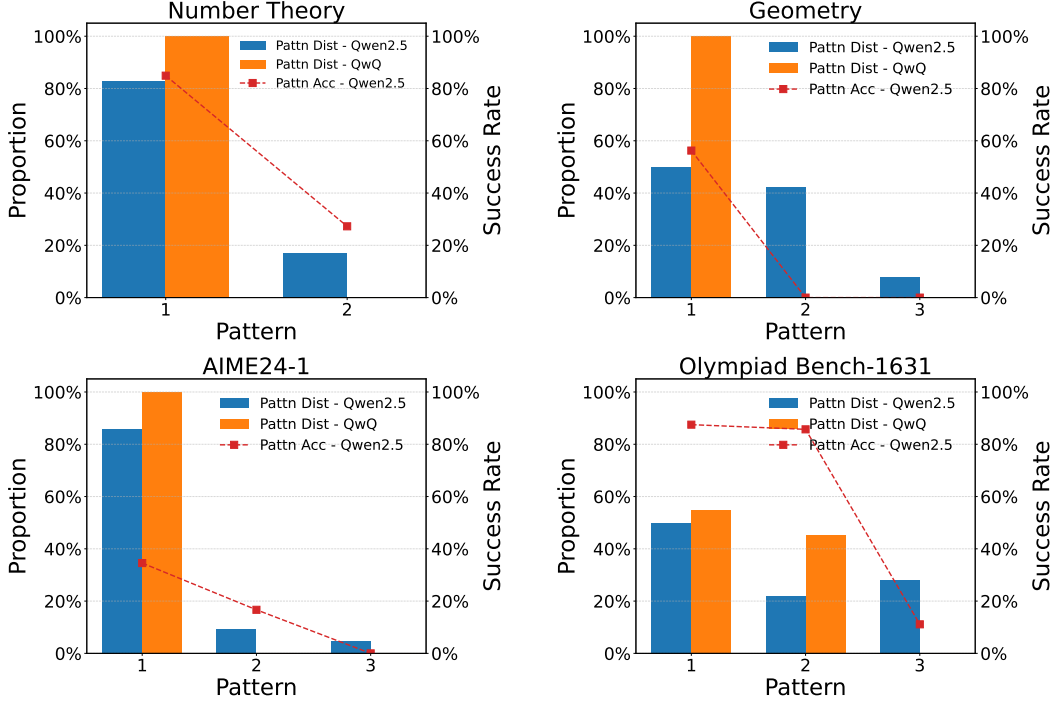


Figure 5: Evaluation results on the larger models Qwen-2.5-32B and QwQ-32B. QwQ’s reasoning patterns converge to the most accurate pattern of Qwen-2.5, demonstrating that our theory continues to apply at larger model scales.

B Experimental Details

B.1 Experimental Setup

Entire Training Procedures: As a representative training setup, we select Qwen2.5-3B as the base model and train it on the MATH dataset using both RLVR and RLIF. We adopt the verl [31] framework and keep all parameter settings identical to the MATH example provided by verl, isolating the effect of the reward. For RLIF, we modify only the reward computation. By varying the random seed, we obtain three training curves for each method, ensuring the reported trends are not artifacts of a single run.

Token-level Analysis: For this experiment we use Qwen2.5-3B together with the RL-enhanced models described above, evaluating on datasets of varying difficulty to probe the generality of the effect. For GSM8K and MATH, we randomly sample 400 questions from their respective test sets; for AIME24, we use the full dataset. For each question we first sample the corresponding answer from the base model, then concatenate this answer to the question and feed it into the RL-enhanced model as a new prompt. Using the interface in the OpenAI library, we compute the probability of each token in the prompt and its rank among the probabilities of all possible tokens at the current position. Comparing whether each token’s rank changes before and after RL yields the proportion of tokens whose ranking shifts.

Reasoning-pattern Level Analysis: Here we conduct two complementary types of experiments. First, we study the dynamics of the pattern distribution during training using our previously trained models, namely Qwen2.5-3B and RL-enhanced Qwen2.5-3B, evaluating every 20 steps. Second, to test how broadly our conclusions hold across different base models and larger scales, we examine the open-source models Qwen-2.5-7B-SimpleRL-Zoo and QwQ-32B together with their corresponding base models. For these released checkpoints we can only compare the pattern distributions of the outputs before and after RL, rather than tracking the pattern-distribution dynamics throughout training.

B.2 Reasoning Patterns Analysis Pipeline

Model Inference: To cover a broad range of problem types and difficulties, we randomly sample questions of different task types (such as Geometry, Algebra, and others) and different difficulty levels (Level 4 and Level 5) from datasets including MATH. For each question, the model samples 1024 answers in the pattern-distribution dynamics task, and between 64 and 1024 answers in the task that considers only the distribution change before and after RL, depending on the task category. These responses are then used for subsequent pattern extraction and classification.

Reasoning Patterns Extraction: We sample a subset of the base model’s responses and use the GPT-4o [19] API to summarize the pattern categories from these samples. This extraction procedure is task-agnostic and applies uniformly to every task we study; the full prompt used for reasoning pattern extraction is given in Appendix D.1. To ensure consistent extraction, we set the temperature of GPT-4o’s API to 0.

Responses Classification: We again employ GPT-4o to classify all responses according to the identified pattern categories, again setting the temperature to 0. The full prompt used for response classification is given in Appendix D.2.

Accuracy Analysis: To isolate the impact of the pattern distribution on accuracy from any change in the reasoning process itself, we construct new prompts by concatenating the original prompt with partial responses from the base model, assuming these partial responses sufficiently represent specific reasoning patterns. Fixing the reasoning process in this way lets us attribute changes in model accuracy to the distribution over reasoning patterns rather than to the reasoning process. Concretely, we remove the sentence containing the final answer from the base model’s response, append the preceding sentence to the end of the question, and feed this as a new prompt to the RL-enhanced model. The RL-enhanced model then continues to predict the answer, and we take the accuracy of this predicted answer as the accuracy of the reasoning pattern.

B.3 Details for Each Task

To convey the breadth of settings our study covers, we detail below the experimental procedures for each task, including task descriptions and representative examples of reasoning patterns (Table 2). Example task prompts are provided in Appendix D.3.

Number Theory: This task presents models with problems involving the distribution of coins across multiple bags. Initially, the bags contain equal numbers of coins; after receiving additional coins and redistributing them equally, the total must exceed a specified value while maintaining equal distribution, and the model must determine the minimum initial coin count per bag. We derived this task from a level 5 MATH problem (id: test/number_theory/1055) [16], creating 32 variants by adjusting parameters such as the bag count and coin totals to form our evaluation dataset. We employ one-shot prompting with a simple, unrelated example that guides answer formatting via "boxed" notation without influencing the reasoning approach. For evaluation, we first choose 20 questions from the evaluation dataset with 4 responses each for reasoning pattern extraction, sampling 64 responses per question. We then categorize all responses by the extracted patterns, which lets us compare how the distribution of the model’s reasoning patterns shifts before and after RL.

Geometry: For the geometry task, the model is given the lengths of two sides of a scalene triangle and asked how many distinct integer centimeter lengths the third side can take. This task is derived from a level 4 geometry problem in MATH (id: test/geometry/1046). Our evaluation follows exactly the same pipeline as for Number Theory, underscoring that the methodology transfers unchanged across domains: we construct a synthetic dataset of 32 questions, sample 20 questions and select 4 responses each from the base model (Qwen2.5-7b-Instruct) for reasoning pattern extraction, and then compare the responses of the model with and without RLVR enhancement over all questions in the evaluation dataset, using 64 responses each for pattern analysis.

Algebra: The algebra task is a can-recycling problem: starting from a certain number of old cans, where a predefined n old cans can be recycled into a new can, the model must determine how many cans can ultimately be produced. The difficulty is that at each iterative step there may be leftover

Task	Reasoning Patterns Examples	Common Elements / Key Words
Number Theory	Modular Congruence with Coefficient Simplification	Uses modular arithmetic to express divisibility conditions.
	Inequality-Driven Search for Minimal Solution	Often involves substituting back to compute the total coins.
Geometry	Systematic Inequality Application	Explicitly lists and solves each of the three inequalities.
	Verification of Scalene Condition	Adding an extra layer of validation to verifies the third side length.
Algebra	Iterative Floor Division with Leftover Tracking.	Uses floor division to compute new cans per step.
	Recursive Recycling with Aggregated Leftovers.	Combines leftovers with newly produced cans before recycling.
Calculus	Direct Simplification and Principal Value Matching	Assumes inputs fall within the principal range.
	Interval Analysis with Case Splitting	Splits the whole domain into different intervals.
Counting and Probability	Direct Probability Setup and Quadratic Solution	Explicitly calculates combinations for total and favorable outcomes.
	Early Simplification and Cross-Multiplication	Early cross-multiplication to eliminate denominators.
AMC23-Q19	Prime Factorization and Simplification	Counting Digits in Large Numbers.
	Rewriting (8^5) as (2^{15}) (15^5) as $(3^5 \cdot 5^5)$.	Final step of counting digits: $(3 \text{ (from 243)} + 15 \text{ (zeros)} = 18)$.
AIME24-Q1	Direct Equation Setup and Elimination	Straightforward and relies on algebraic manipulation.
	Alternative Equation Formulation and Solving	Expressing variables in terms of others early on.
OlympiadBench-Q1631	Direct Calculation and Empirical Testing	Relying on direct computation and empirical verification.
	Factorization-Based Reasoning	Using algebraic factorization to argue certain terms y_n .

Table 2: Representative reasoning pattern examples spanning the full range of tasks and domains in our study.

cans that cannot be divided evenly, and these may eventually be combined for further recycling. The template for this task comes from a level 4 algebra problem in MATH (id: test/algebra/2768). For this and all subsequent supplementary tasks, we adopt the same settings as the previous tasks—including dataset size, number of samples, and so on—so that results remain directly comparable across the full suite of domains.

Calculus: Our calculus task is deliberately simple and self-contained: we present a single-variable trigonometric equation together with the domain of the variable and ask the model to determine the number of solutions. Despite its simplicity, this task probes the model’s basic competence on calculus-style problems and thereby broadens the range of domains our analysis covers. Its prototype is a level 4 pre-calculus problem in MATH (id: test/precalculus/1140).

Counting and Probability: We further study model performance on a counting and probability task. Given a collection of white and black balls, two balls are drawn at random and the probability that one is black and the other white is specified; provided the number of balls of one color, the model must compute the minimum number of balls of the other color. This probability-theory task, which extends our study into yet another mathematical domain, comes from a level 4 counting and probability problem in MATH (id: test/counting_and_probability/79).

AMC23 Question: To extend our evaluation beyond the MATH-derived tasks to a competition benchmark, we use question 19 from AMC23 [1]. For reasoning pattern extraction we randomly sample 48 responses, and we then sample 1024 responses for reasoning pattern analysis.

690 B.4 Details for Experiments on QwQ-32B and Qwen-2.5-32B

691 To probe generality at large scale, we conduct experiments on four tasks that span multiple bench-
 692 marks. Because of the constraints of API-based testing, we randomly select one question per task
 693 as input, sample 64 responses from each of the two models, and directly perform reasoning pattern
 694 extraction and classification on these answers. Specifically, from the MATH dataset we choose
 695 question 1055 (number theory) and question 1046 (geometry); for the remaining two tasks we use
 696 problem 1 from AIME24 and question 1631 from OlympiadBench. Examples of the resulting patterns
 697 are shown in Table 2.

698 C Deferred Proofs

699 In this appendix, we provide proofs for our main theoretical results: Proposition 5.2, Theorems 5.3,
 700 Theorems 5.4 and Theorems 5.6.

701 C.1 Proof of Proposition 5.2

702 *Proof of Proposition 5.2.* In this proof, we utilize the proof techniques in [26], recall that the opti-
 703 mization objective of RL is

$$\phi_{\text{RL}}(\theta) = \mathbb{E}_{\mathbf{x} \sim \mathcal{D}, \mathbf{y} \sim \pi_{\theta}(\mathbf{y}|\mathbf{x})} [r_{\phi}(\mathbf{x}, \mathbf{y})] - \beta \mathbb{D}_{\text{KL}} [\pi_{\theta}(\mathbf{y} | \mathbf{x}) \parallel \pi_{\text{ref}}(\mathbf{y} | \mathbf{x})], \quad (\text{C.1})$$

704 Under our framework, as state in section 5.1, we consider a policy conduct reasoning by first sample
 705 $\mathbf{r}_i \in \mathcal{R}$ based on $\pi_{\theta}(\mathbf{r}_i|\mathbf{q})$ and then provide the final answer $\mathbf{a} \in \mathcal{A}$ by $\pi_{\theta}(\mathbf{a}|\mathbf{r}_i)$, the reward $\phi_{\text{RL}}(\theta)$
 706 can be written as

$$\phi_{\text{RL}}(\theta) = \mathbb{E}_{\mathbf{y} \sim \pi_{\theta}(\mathbf{y}|\mathbf{q})} \left[\sum_{\mathbf{a} \in \mathcal{A}} \pi_{\text{ref}}(\mathbf{a} | \mathbf{r}) r_{\phi}([\mathbf{r}, \mathbf{a}], \mathbf{q}) \right] - \beta \mathbb{D}_{\text{KL}} [\pi_{\theta}(\mathbf{y} | \mathbf{x}) \parallel \pi_{\text{ref}}(\mathbf{y} | \mathbf{x})], \quad (\text{C.2})$$

707 Let $R(\mathbf{r}) = \sum_{\mathbf{a} \in \mathcal{A}} \pi_{\text{ref}}(\mathbf{a} | \mathbf{r}) r_{\phi}([\mathbf{r}, \mathbf{a}], \mathbf{q})$, then

$$\begin{aligned} \phi_{\text{RL}}(\theta) &= \mathbb{E}_{\mathbf{r} \sim \pi_{\theta}(\cdot|\mathbf{q}), \mathbf{a} \sim \pi_{\theta}(\cdot|\mathbf{r})} \left[R(\mathbf{r}) - \beta \ln \left(\frac{\pi_{\theta}(\mathbf{r}|\mathbf{q}) \cdot \pi_{\text{ref}}(\mathbf{a}|\mathbf{r})}{\pi_{\text{ref}}(\mathbf{r}|\mathbf{q}) \cdot \pi_{\text{ref}}(\mathbf{a}|\mathbf{r})} \right) \right] \\ &= \mathbb{E}_{\mathbf{r} \sim \pi_{\theta}(\cdot|\mathbf{q})} \left[R(\mathbf{r}) - \beta \ln \left(\frac{\pi_{\theta}(\mathbf{r}|\mathbf{q})}{\pi_{\text{ref}}(\mathbf{r}|\mathbf{q})} \right) \right] \\ &= -\beta \mathbb{E}_{\mathbf{r} \sim \pi_{\theta}(\cdot|\mathbf{q})} \left[\ln \left(\frac{\pi_{\theta}(\mathbf{r}|\mathbf{q})}{\frac{1}{Z} \exp \left(\frac{1}{\beta} R(\mathbf{r}) \right) \pi_{\text{ref}}(\mathbf{r}|\mathbf{q})} \right) - \ln Z \right] \\ &= -\beta \mathbb{D}_{\text{KL}} [\pi^*(\mathbf{r}|\mathbf{q}) \parallel \pi_{\text{ref}}(\mathbf{y} | \mathbf{x})] + \beta \ln Z. \end{aligned}$$

708 Where the third equation is by Assumption 5.1 and $Z = \sum_{\mathbf{r} \in \mathcal{R}} \exp \left(\frac{1}{\beta} p^*(\mathbf{r}) \right) \pi_{\theta_{\text{ref}}}(\mathbf{r}|\mathbf{q})$ is the
 709 partition constant that ensures

$$\pi^*(\mathbf{r}|\mathbf{q}) = \frac{1}{Z} \exp \left(\frac{1}{\beta} R(\mathbf{r}) \right) \pi_{\theta}(\mathbf{r}|\mathbf{q}),$$

710 is a valid probability distribution such that $\sum_{\mathbf{r} \in \mathcal{R}} \pi^*(\mathbf{r}|\mathbf{q}) = 1$. Since Z is not a function of $\mathbf{r}$.

711 Therefore, maximizing the objective in Equation C.2 is equivalent to:

$$\begin{aligned} &\max_{\pi} \mathbb{E}_{\mathbf{x} \sim \mathcal{D}, \mathbf{y} \sim \pi(\mathbf{y}|\mathbf{x})} [r_{\phi}(\mathbf{x}, \mathbf{y})] - \beta \mathbb{D}_{\text{KL}} [\pi(\mathbf{y} | \mathbf{x}) \parallel \pi_{\text{ref}}(\mathbf{y} | \mathbf{x})] \\ &= \min_{\pi} \beta \mathbb{D}_{\text{KL}} [\pi^*(\mathbf{r}|\mathbf{q}) \parallel \pi_{\text{ref}}(\mathbf{y} | \mathbf{x})] - \beta \ln Z. \end{aligned}$$

712 By the properties of KL-divergence, we know that the optimal policy for the KL-constrained reward
 713 maximization objective satisfies:

$$\pi_{\text{opt}}(\mathbf{r}|\mathbf{q}) = \frac{1}{Z} \exp \left(\frac{1}{\beta} R(\mathbf{r}) \right) \pi_{\theta_{\text{ref}}}(\mathbf{r}|\mathbf{q}) \text{ for all } \mathbf{r} \in \mathcal{R}.$$

specifically, for RLVR, $R_{\text{RLVR}}(\mathbf{r}) = p^*$, for RLIF, we have

$$\begin{aligned} R'(\mathbf{r}) &= - \left(\frac{1}{|\mathcal{R}|} \sum_{\mathbf{r}_i \in \mathcal{R}} \log(|\mathcal{R}| \cdot \pi_{\theta}(\mathbf{r}_i|q)) + \frac{1}{|\mathcal{A}|} \sum_{\mathbf{a}_j \in \mathcal{A}} \log(|\mathcal{A}| \cdot \pi_{\theta}(\mathbf{a}_j|q, \mathbf{r})) \right) \\ &= - \frac{1}{|\mathcal{A}|} \sum_{\mathbf{a}_j \in \mathcal{A}} \log(\pi_{\theta}(\mathbf{a}_j|q, \mathbf{r})) - \underbrace{\left(\frac{1}{|\mathcal{R}|} \sum_{\mathbf{r}_i \in \mathcal{R}} \log(|\mathcal{R}| \cdot \pi_{\theta}(\mathbf{r}_i|q)) + \log|\mathcal{A}| \right)}_c, \end{aligned}$$

where the second term is the same for all $\mathbf{r} \in |\mathcal{R}|$, for RLIF, the optimal policy can be written as

$$\begin{aligned} \pi_{\text{opt}}(\mathbf{r}|q) &= \frac{1}{Z} \exp\left(\frac{1}{\beta} R'(\mathbf{r})\right) \pi_{\theta_{\text{ref}}}(\mathbf{r}|q) \\ &= \frac{\exp\left(\frac{1}{\beta} R'(\mathbf{r})\right) \pi_{\theta_{\text{ref}}}(\mathbf{r}|q)}{\sum_{\mathbf{r}_i \in \mathcal{R}} \exp\left(\frac{1}{\beta} (R'(\mathbf{r}_i))\right) \pi_{\theta_{\text{ref}}}(\mathbf{r}_i|q)} \\ &= \frac{\exp\left(\frac{1}{\beta} (R'(\mathbf{r}) - c)\right) \pi_{\theta_{\text{ref}}}(\mathbf{r}|q)}{\sum_{\mathbf{r}_i \in \mathcal{R}} \exp\left(\frac{1}{\beta} (R'(\mathbf{r}_i) - c)\right) \pi_{\theta_{\text{ref}}}(\mathbf{r}_i|q)} \end{aligned}$$

So we can write $R_{\text{RLIF}}(\mathbf{r}) = R'(\mathbf{r}) - c = -\frac{1}{|\mathcal{A}|} \sum_{\mathbf{a}_j \in \mathcal{A}} \log(\pi_{\theta}(\mathbf{a}_j|q, \mathbf{r}))$. This concludes the proof of the theorem. $\square$

C.2 Dynamics of Tabular Policy

Consider the RL objective (Eq 3.1), using a tabular policy (Eq 5.3) with Assumption 5.1 holds, the gradient dynamics of θ can be computed as

$$\begin{aligned} \frac{d}{dt} \theta(t) &= \nabla \phi_{\text{RL}}(\theta(t)) \\ &= \nabla \mathbb{E}_{\mathbf{r} \sim \pi_{\theta}(\cdot|q)} [r_{\phi}^{KL}(\mathbf{r}|q)] \\ &= \sum_{\mathbf{r} \in \mathcal{R}} [r_{\phi}^{KL}(\mathbf{r}|q) \nabla \pi_{\theta(t)}(\mathbf{r}|q)], \end{aligned}$$

Where $\pi_{\theta(t)}(\mathbf{r}|q) = \text{softmax}(\theta_{:,q})_{\mathbf{r}}$, $\theta_{:,q} \in \mathbb{R}^{|\mathcal{V}|}$ is the column of θ corresponding to q , and $r_{\phi}(\mathbf{r}|q)^{KL}$ refers to the reward with KL divergence, so

$$\begin{aligned} \frac{\partial \theta_{\mathbf{r}_i, q}(t)}{\partial t} &= \sum_{\mathbf{r}_j \in \mathcal{R}} [r_{\phi}^{KL}(\mathbf{r}_j|q) (\nabla \text{softmax}(\theta_{:,q})_{\mathbf{r}_j})_{\mathbf{r}_i}] \\ &= \sum_{\mathbf{r}_j \in \mathcal{R}} [r_{\phi}^{KL}(\mathbf{r}_j|q) (-\pi_{\theta(t)}(\mathbf{r}_j|q) \cdot \pi_{\theta(t)}(\mathbf{r}_i|q))] \\ &\quad + r_{\phi}^{KL}(\mathbf{r}_i|q) \cdot \pi_{\theta(t)}(\mathbf{r}_i|q) \end{aligned}$$

where the overall accuracy is defined as $\text{ACC}_{\theta(t)} = \sum_{\mathbf{r} \in \mathcal{R}} \pi_{\theta(t)}(\mathbf{r}|q) p^*(\mathbf{r})$. For $\pi_{\theta(t)}(\mathbf{r}_i|q)$, we have

$$\frac{d}{dt} \pi_{\theta(t)}(\mathbf{r}_i|q) = \sum_{\mathbf{r}_j \in \mathcal{R}} \frac{\partial \pi_{\theta(t)}(\mathbf{r}_i|q)}{\partial \theta_{\mathbf{r}_j, q}(t)} \cdot \frac{\partial \theta_{\mathbf{r}_j, q}(t)}{\partial t}$$

C.3 Proof of Theorem 5.3

To proof Theorem 5.3, here we first prove that $\text{ACC}_{\theta(t)} \geq \text{ACC}_{\theta(0)}$ for $t \geq 0$ (Eq C.4), then we establish a lower bound for $\pi_{\theta(t)}(\mathbf{r}^*|q)$ (Eq C.5), finally we derive the final bound for t stated in Theorem 5.3.

729 *Proof of Theorem 5.3.* Setting $\beta = 0$, Eq C.3 becomes:

$$\frac{\partial \theta_{\mathbf{r}_i, \mathbf{q}}(t)}{\partial t} = \pi_{\theta(t)}(\mathbf{r}_i | \mathbf{q}) \cdot (p^*(\mathbf{r}_i) - \text{ACC}_{\theta(t)}). \quad (\text{C.3})$$

730 For $\pi_{\theta(t)}(\mathbf{r}_i | \mathbf{q})$, we derive:

$$\begin{aligned} \frac{d}{dt} \pi_{\theta(t)}(\mathbf{r}_i | \mathbf{q}) &= \sum_{\mathbf{r}_j \in \mathcal{R}} \frac{\partial \pi_{\theta(t)}(\mathbf{r}_i | \mathbf{q})}{\partial \theta_{\mathbf{r}_j, \mathbf{q}}(t)} \cdot \frac{\partial \theta_{\mathbf{r}_j, \mathbf{q}}(t)}{\partial t} \\ &= \sum_{\mathbf{r}_j \in \mathcal{R}} (\text{ACC}_{\theta(t)} - p^*(\mathbf{r}_j)) \pi_{\theta(t)}^2(\mathbf{r}_j | \mathbf{q}) \pi_{\theta(t)}(\mathbf{r}_i | \mathbf{q}) \\ &\quad + (p^*(\mathbf{r}_i) - \text{ACC}_{\theta(t)}) \pi_{\theta(t)}^2(\mathbf{r}_i | \mathbf{q}), \end{aligned}$$

731 and for the accuracy $\text{ACC}_{\theta(t)}$:

$$\begin{aligned} \frac{d}{dt} \text{ACC}_{\theta(t)} &= \sum_{\mathbf{r}_i \in \mathcal{R}} p^*(\mathbf{r}_i) \cdot \frac{d}{dt} \pi_{\theta(t)}(\mathbf{r}_i | \mathbf{q}) \\ &= \sum_{\mathbf{r}_i, \mathbf{r}_j \in \mathcal{R}} p^*(\mathbf{r}_i) \cdot (\text{ACC}_{\theta(t)} - p^*(\mathbf{r}_j)) \pi_{\theta(t)}^2(\mathbf{r}_j | \mathbf{q}) \pi_{\theta(t)}(\mathbf{r}_i | \mathbf{q}) \\ &\quad + \sum_{\mathbf{r}_i \in \mathcal{R}} (p^*(\mathbf{r}_i) - \text{ACC}_{\theta(t)}) \pi_{\theta(t)}^2(\mathbf{r}_i | \mathbf{q}) \\ &= \sum_{\mathbf{r}_j \in \mathcal{R}} \text{ACC}_{\theta(t)} \cdot (\text{ACC}_{\theta(t)} - p^*(\mathbf{r}_j)) \pi_{\theta(t)}^2(\mathbf{r}_j | \mathbf{q}) \\ &\quad + \sum_{\mathbf{r}_i \in \mathcal{R}} (p^*(\mathbf{r}_i) - \text{ACC}_{\theta(t)}) \pi_{\theta(t)}^2(\mathbf{r}_i | \mathbf{q}) p^*(\mathbf{r}_i) \\ &= \sum_{\mathbf{r}_i \in \mathcal{R}} \pi_{\theta(t)}^2(\mathbf{r}_i | \mathbf{q}) (p^*(\mathbf{r}_i) - \text{ACC}_{\theta(t)})^2 \geq 0. \end{aligned} \quad (\text{C.4})$$

732 As $\frac{d}{dt} \text{ACC}_{\theta(t)} \geq 0$, so $\text{ACC}_{\theta(t)} > \text{ACC}_{\theta(0)}$ holds for $t \geq 0$, for the optimal reasoning pattern
733 $\mathbf{r}^*$, let $\mathbf{r}'$ the suboptimal reasoning pattern with the second highest success rate, defined as $\mathbf{r}' =$
734 $\arg \max_{\mathbf{r}, \mathbf{r} \neq \mathbf{r}^*} p^*(\mathbf{r})$, we have:

$$\begin{aligned} \frac{d}{dt} \pi_{\theta(t)}(\mathbf{r}^* | \mathbf{q}) &= \sum_{\mathbf{r}_j \in \mathcal{R}} (\text{ACC}_{\theta(t)} - p^*(\mathbf{r}_j)) \pi_{\theta(t)}^2(\mathbf{r}_j | \mathbf{q}) \pi_{\theta(t)}(\mathbf{r}^* | \mathbf{q}) \\ &\quad + (p^*(\mathbf{r}^*) - \text{ACC}_{\theta(t)}) \pi_{\theta(t)}^2(\mathbf{r}^* | \mathbf{q}) \\ &\geq (\text{ACC}_{\theta(t)} - p^*(\mathbf{r}^*)) \pi_{\theta(t)}^2(\mathbf{r}^* | \mathbf{q}) \pi_{\theta(t)}(\mathbf{r}^* | \mathbf{q}) \\ &\quad + (p^*(\mathbf{r}^*) - \text{ACC}_{\theta(t)}) \pi_{\theta(t)}^2(\mathbf{r}^* | \mathbf{q}) \\ &= (p^*(\mathbf{r}^*) - \text{ACC}_{\theta(t)}) \pi_{\theta(t)}^2(\mathbf{r}^* | \mathbf{q}) (1 - \pi_{\theta(t)}(\mathbf{r}^* | \mathbf{q})) \\ &\geq (p^*(\mathbf{r}^*) - p^*(\mathbf{r}')) \pi_{\theta(t)}^2(\mathbf{r}^* | \mathbf{q}) (1 - \pi_{\theta(t)}(\mathbf{r}^* | \mathbf{q}))^2 \geq 0. \end{aligned} \quad (\text{C.5})$$

735 As $\frac{d}{dt} \pi_{\theta(t)}(\mathbf{r}^* | \mathbf{q}) \geq 0$, $\pi_{\theta(t)}(\mathbf{r}^* | \mathbf{q}) \geq \pi_{\theta(0)}(\mathbf{r}^* | \mathbf{q})$ holds for $t \geq 0$, define $C =$
736 $(p^*(\mathbf{r}^*) - p^*(\mathbf{r}')) \pi_{\theta(0)}^2(\mathbf{r}^* | \mathbf{q})$:

$$\frac{d}{dt} \pi_{\theta(t)}(\mathbf{r}^* | \mathbf{q}) \geq C (1 - \pi_{\theta(t)}(\mathbf{r}^* | \mathbf{q}))^2.$$

737 This differential inequality yields:

$$\pi_{\theta(t)}(\mathbf{r}^* | \mathbf{q}) > 1 - \frac{1}{Ct + \frac{1}{1 - \pi_{\theta(0)}(\mathbf{r}^* | \mathbf{q})}}.$$

738 For any $\epsilon > 0$, there exists $T_1 = \frac{1}{C} \left(\frac{1}{\epsilon} - \frac{1}{1 - \pi_{\theta(0)}(\mathbf{r}^* | \mathbf{q})} \right) = \mathcal{O}(\frac{1}{\epsilon})$, such that

$$\pi_{\theta(t)}(\mathbf{r}^* | \mathbf{q}) > 1 - \epsilon.$$

739 This concludes the proof of the theorem. $\square$

740 **C.4 Proof of Theorem 5.4**

741 To proof Theorem 5.4, we first establish an upper bound for $\pi_{\theta(t)}(\mathbf{r}'|\mathbf{q})$ (Eq C.6), then we analyze
 742 the ratio $\rho_{\theta(t)}(\mathbf{r}_i) = \frac{\pi_{\theta(t)}(\mathbf{r}_i|\mathbf{q})}{\pi_{\theta(t)}(\mathbf{r}^*|\mathbf{q})}$ (Eq C.7) and prove that for $t > T_0$, $\text{ACC}_{\theta(t)} \geq p^*(\mathbf{r}')$.

743 *Proof of Theorem 5.4.* Let $\mathbf{r}^*$ and $\mathbf{r}'$ be the optimal and second optimal reasoning patterns. For
 744 $\pi_{\theta(t)}(\mathbf{r}'|\mathbf{q})$:

$$\begin{aligned}
 \frac{d}{dt} \pi_{\theta(t)}(\mathbf{r}'|\mathbf{q}) &= \sum_{\mathbf{r}_j \in \mathcal{R}} (\text{ACC}_{\theta(t)} - p^*(\mathbf{r}_j)) \pi_{\theta(t)}^2(\mathbf{r}_j|\mathbf{q}) \pi_{\theta(t)}(\mathbf{r}'|\mathbf{q}) \\
 &\quad + (p^*(\mathbf{r}') - \text{ACC}_{\theta(t)}) \pi_{\theta(t)}^2(\mathbf{r}'|\mathbf{q}) \\
 &= \sum_{\mathbf{r}_j \in \mathcal{R}/\{\mathbf{r}^*\}} (\text{ACC}_{\theta(t)} - p^*(\mathbf{r}_j)) \pi_{\theta(t)}^2(\mathbf{r}_j|\mathbf{q}) \pi_{\theta(t)}(\mathbf{r}'|\mathbf{q}) \\
 &\quad + (p^*(\mathbf{r}') - \text{ACC}_{\theta(t)}) \pi_{\theta(t)}^2(\mathbf{r}'|\mathbf{q}) (1 - \pi_{\theta(t)}(\mathbf{r}^*|\mathbf{q})) \\
 &\leq \sum_{\mathbf{r}_j \in \mathcal{R}/\{\mathbf{r}^*\}} \pi_{\theta(t)}^2(\mathbf{r}_j|\mathbf{q}) \pi_{\theta(t)}(\mathbf{r}'|\mathbf{q}) \\
 &\quad + (p^*(\mathbf{r}') - p^*(\mathbf{r}') \pi_{\theta(t)}(\mathbf{r}^*|\mathbf{q})) \pi_{\theta(t)}^2(\mathbf{r}'|\mathbf{q}) (1 - \pi_{\theta(t)}(\mathbf{r}^*|\mathbf{q})) \\
 &\leq \pi_{\theta(t)}(\mathbf{r}'|\mathbf{q}) (1 - \pi_{\theta(t)}(\mathbf{r}^*|\mathbf{q}))^2 + p^*(\mathbf{r}') \pi_{\theta(t)}^2(\mathbf{r}'|\mathbf{q}) (1 - \pi_{\theta(t)}(\mathbf{r}^*|\mathbf{q}))^2 \\
 &\leq 2 (1 - \pi_{\theta(t)}(\mathbf{r}^*|\mathbf{q}))^2.
 \end{aligned}$$

745 This yields:

$$\pi_{\theta(t)}(\mathbf{r}'|\mathbf{q}) \leq 1 - \frac{1}{2t + \frac{1}{1 - \pi_{\theta(0)}(\mathbf{r}'|\mathbf{q})}}. \quad (\text{C.6})$$

746 Next, define $\rho_{\theta(t)}(\mathbf{r}_i) = \frac{\pi_{\theta(t)}(\mathbf{r}_i|\mathbf{q})}{\pi_{\theta(t)}(\mathbf{r}^*|\mathbf{q})}$, then for all $\mathbf{r}_i \in \mathcal{R}/\{\mathbf{r}^*, \mathbf{r}'\}$, we have

$$\begin{aligned}
 \frac{d}{dt} \rho_{\theta(t)}(\mathbf{r}_i) &= \frac{\pi_{\theta(t)}(\mathbf{r}^*|\mathbf{q}) \frac{d}{dt} \pi_{\theta(t)}(\mathbf{r}_i|\mathbf{q}) - \pi_{\theta(t)}(\mathbf{r}_i|\mathbf{q}) \frac{d}{dt} \pi_{\theta(t)}(\mathbf{r}^*|\mathbf{q})}{\pi_{\theta(t)}^2(\mathbf{r}^*|\mathbf{q})} \\
 &= \frac{\pi_{\theta(t)}(\mathbf{r}_i|\mathbf{q})}{\pi_{\theta(t)}(\mathbf{r}^*|\mathbf{q})} ((p^*(\mathbf{r}_i) - \text{ACC}_{\theta(t)}) \pi_{\theta(t)}(\mathbf{r}_i|\mathbf{q}) - (p^*(\mathbf{r}^*) - \text{ACC}_{\theta(t)}) \pi_{\theta(t)}(\mathbf{r}^*|\mathbf{q})) \\
 &\leq -\rho_{\theta(t)}(\mathbf{r}_i) (p^*(\mathbf{r}^*) - \text{ACC}_{\theta(t)}) \pi_{\theta(t)}(\mathbf{r}^*|\mathbf{q}) \leq 0
 \end{aligned}$$

747 The last inequality is based on the condition for case 2 and $\text{ACC}_{\theta(t)} > \text{ACC}_{\theta(0)}$, so $p^*(\mathbf{r}_i) <$
 748 $\text{ACC}_{\theta(t)}$ holds for all $\mathbf{r}_i \in \mathcal{R}/\{\mathbf{r}^*\}$, $t \geq 0$. Then as $\frac{d}{dt} \rho_{\theta(t)}(\mathbf{r}_i) \leq 0$, we have:

$$\pi_{\theta(t)}(\mathbf{r}^*|\mathbf{q}) = \frac{1 - \pi_{\theta(t)}(\mathbf{r}'|\mathbf{q})}{\sum_{\mathbf{r}_i \in \mathcal{R}/\{\mathbf{r}'\}} \rho_{\theta(t)}(\mathbf{r}_i)} \geq \frac{1 - \pi_{\theta(t)}(\mathbf{r}'|\mathbf{q})}{\sum_{\mathbf{r}_i \in \mathcal{R}/\{\mathbf{r}'\}} \rho_{\theta(0)}(\mathbf{r}_i)}$$

749 While $\text{ACC}_{\theta(t)} < p^*(\mathbf{r}')$, define $\gamma_{\pi_{\text{ref}}} := \sum_{\mathbf{r} \in \mathcal{R}/\{\mathbf{r}'\}} \frac{\pi_{\theta_{\text{ref}}}(\mathbf{r}|\mathbf{q})}{\pi_{\theta_{\text{ref}}}(\mathbf{r}^*|\mathbf{q})}$ we have:

$$\begin{aligned}
 \frac{d}{dt} \rho_{\theta(t)}(\mathbf{r}_i) &\leq -\rho_{\theta(t)}(\mathbf{r}_i) (p^*(\mathbf{r}^*) - \text{ACC}_{\theta(t)}) \pi_{\theta(t)}(\mathbf{r}^*|\mathbf{q}) \\
 &\leq -\rho_{\theta(t)}(\mathbf{r}_i) (p^*(\mathbf{r}^*) - p^*(\mathbf{r}')) \frac{1 - \pi_{\theta(t)}(\mathbf{r}'|\mathbf{q})}{\gamma_{\pi_{\text{ref}}}} \\
 &\leq -\rho_{\theta(t)}(\mathbf{r}_i) \frac{p^*(\mathbf{r}^*) - p^*(\mathbf{r}')}{\gamma_{\pi_{\text{ref}}}} \frac{1}{2t + \frac{1}{1 - \pi_{\theta(0)}(\mathbf{r}'|\mathbf{q})}}
 \end{aligned}$$

750 Then we have:

$$\frac{d}{dt} \rho_{\theta(t)}(\mathbf{r}_i) \leq -\rho_{\theta(t)}(\mathbf{r}_i) \frac{p^*(\mathbf{r}^*) - p^*(\mathbf{r}')}{\gamma_{\pi_{\text{ref}}}} \frac{1}{2t + \frac{1}{1 - \pi_{\theta(0)}(\mathbf{r}'|\mathbf{q})}}, \quad (\text{C.7})$$

751 this differential inequality leads to:

$$\rho_{\theta(t)}(\mathbf{r}_i) \leq \frac{(1 - \pi_{\theta(0)}(\mathbf{r}'|\mathbf{q}))^{-\Delta/(2\gamma_{\pi_{\text{ref}}})}}{(1/(1 - \pi_{\theta(0)}(\mathbf{r}'|\mathbf{q})) + 2t)^{\Delta/(2\gamma_{\pi_{\text{ref}}})}} \rho_{\theta(0)}(\mathbf{r}_i).$$

752 Where $\Delta = p^*(\mathbf{r}^*) - p^*(\mathbf{r}')$. Recall that:

$$\begin{aligned} \text{ACC}_{\theta(t)} - p^*(\mathbf{r}') &= \sum_{\mathbf{r} \in \mathcal{R}} \pi_{\theta(t)}(\mathbf{r}|\mathbf{q}) p^*(\mathbf{r}) - p^*(\mathbf{r}') \\ &= p^*(\mathbf{r}^*) \left[(p^*(\mathbf{r}^*) - p^*(\mathbf{r}')) + \sum_{\mathbf{r}_i \in \mathcal{R}/\{\mathbf{r}'\}} \rho_{\theta(t)}(\mathbf{r}_i) (p^*(\mathbf{r}_i) - p^*(\mathbf{r}')) \right] \\ &\geq p^*(\mathbf{r}^*) \left[(p^*(\mathbf{r}^*) - p^*(\mathbf{r}')) - \sum_{\mathbf{r}_i \in \mathcal{R}/\{\mathbf{r}'\}} \rho_{\theta(t)}(\mathbf{r}_i) p^*(\mathbf{r}') \right]. \end{aligned}$$

753 Let $C_2 = 1/\Delta$, $C_1 = p^*(\mathbf{r}')/\Delta$, define:

$$T_0 = \frac{1}{2 - 2\pi_{\theta_{\text{ref}}}(\mathbf{r}'|\mathbf{q})} \left((C_1 \cdot \gamma_{\pi_{\text{ref}}})^{2C_2 \cdot \gamma_{\pi_{\text{ref}}}} - 1 \right),$$

754 then:

$$\begin{aligned} &\sum_{\mathbf{r}_i \in \mathcal{R}/\{\mathbf{r}'\}} \rho_{\theta(T_0)}(\mathbf{r}_i) p^*(\mathbf{r}') \\ &\leq \frac{(1 - \pi_{\theta(0)}(\mathbf{r}'|\mathbf{q}))^{-\Delta/(2\gamma_{\pi_{\text{ref}}})}}{(1/(1 - \pi_{\theta(0)}(\mathbf{r}'|\mathbf{q})) + 2T_0)^{\Delta/(2\gamma_{\pi_{\text{ref}}})}} \sum_{\mathbf{r}_i \in \mathcal{R}/\{\mathbf{r}'\}} \rho_{\theta(0)}(\mathbf{r}_i) p^*(\mathbf{r}') \\ &\leq \frac{(1 - \pi_{\theta(0)}(\mathbf{r}'|\mathbf{q}))^{-\Delta/(2\gamma_{\pi_{\text{ref}}})}}{\left(1/(1 - \pi_{\theta(0)}(\mathbf{r}'|\mathbf{q})) \cdot \left(\frac{\gamma_{\pi_{\text{ref}}} p^*(\mathbf{r}')}{\Delta} \right)^{2\gamma_{\pi_{\text{ref}}}/\Delta} \right)^{\Delta/(2\gamma_{\pi_{\text{ref}}})}} p^*(\mathbf{r}') \sum_{\mathbf{r}_i \in \mathcal{R}/\{\mathbf{r}'\}} \rho_{\theta(0)}(\mathbf{r}_i) \\ &= \frac{1}{\left(\left(\frac{\gamma_{\pi_{\text{ref}}} p^*(\mathbf{r}')}{\Delta} \right)^{2\gamma_{\pi_{\text{ref}}}/\Delta} \right)^{\Delta/(2\gamma_{\pi_{\text{ref}}})}} p^*(\mathbf{r}') \gamma_{\pi_{\text{ref}}} \\ &= \Delta = p^*(\mathbf{r}^*) - p^*(\mathbf{r}'). \end{aligned}$$

755 Finally, for $t > T_0$ we have:

$$\begin{aligned} \text{ACC}_{\theta(t)} - p^*(\mathbf{r}') &\geq p^*(\mathbf{r}^*) \left[(p^*(\mathbf{r}^*) - p^*(\mathbf{r}')) - \sum_{\mathbf{r}_i \in \mathcal{R}/\{\mathbf{r}'\}} \rho_{\theta(t)}(\mathbf{r}_i) p^*(\mathbf{r}') \right] \\ &\geq p^*(\mathbf{r}^*) [(p^*(\mathbf{r}^*) - p^*(\mathbf{r}')) - (p^*(\mathbf{r}^*) - p^*(\mathbf{r}'))] = 0. \end{aligned}$$

756 This concludes the proof of the theorem. $\square$

757 C.5 Proof of Theorem 5.6

758 *Proof of Theorem 5.6.* First, as we consider the case $|\mathcal{A}| = 2$ and $\pi_{\theta_{\text{base}}}(\mathbf{r}|\mathbf{q}) = \frac{1}{|\mathcal{R}|}$, Assumption
759 5.5 holds means $\frac{1}{|\mathcal{R}|} \sum_{\mathbf{r} \in \mathcal{R}} (p^*(\mathbf{r})) > 0.5$. The derivative of overall accuracy can be written as

$$\begin{aligned} \frac{d}{dt} \text{ACC}_{\theta(t)} &= \sum_{\mathbf{r}_i \in \mathcal{R}} p^*(\mathbf{r}_i) \cdot \frac{d}{dt} \pi_{\theta(t)}(\mathbf{r}_i|\mathbf{q}) \\ &= \sum_{\mathbf{r}_i \in \mathcal{R}} \pi_{\theta(t)}^2(\mathbf{r}_i|\mathbf{q}) (p^*(\mathbf{r}_i) - \text{ACC}_{\theta(t)}) \left(- \sum_{\mathbf{a} \in \mathcal{A}} \log(\pi_{\theta(t)}(\mathbf{a}|\mathbf{r}_i, \mathbf{q})) - \text{Con}_{\theta(t)} \right), \end{aligned}$$

760 where $\text{Con}_{\theta(t)} = -\sum_{\mathbf{r}_i \in \mathcal{R}} \pi_{\theta(t)}(\mathbf{r}_i | \mathbf{q}) \sum_{\mathbf{a} \in \mathcal{A}} \log(\pi_{\theta(t)}(\mathbf{a} | \mathbf{r}_i, \mathbf{q}))$. Here we assume $\mathcal{A} = \{\mathbf{a}^*, \mathbf{a}'\}$
 761 where $\mathbf{a}'$ is the incorrect answer, and $\pi_{\theta(0)}(\mathbf{r}_i | \mathbf{q}) = \frac{1}{|\mathcal{R}|}$ for all $\mathbf{r}_i \in \mathcal{R}$. Then:

$$\begin{aligned} \left. \frac{d}{dt} \text{ACC}_{\theta(t)} \right|_{t=0} &= \sum_{\mathbf{r}_i \in \mathcal{R}} \frac{1}{|\mathcal{R}|^2} \left(p^*(\mathbf{r}_i) - \frac{1}{|\mathcal{R}|} \sum_{\mathbf{r}_j \in \mathcal{R}} p^*(\mathbf{r}_j) \right) \cdot \\ &\quad \left(-\sum_{\mathbf{a} \in \mathcal{A}} \log(\pi_{\theta(0)}(\mathbf{a} | \mathbf{r}_i, \mathbf{q})) + \frac{1}{|\mathcal{R}|} \sum_{\mathbf{r}_j \in \mathcal{R}} \sum_{\mathbf{a} \in \mathcal{A}} \log(\pi_{\theta(0)}(\mathbf{a} | \mathbf{r}_j, \mathbf{q})) \right) \\ &= \frac{1}{|\mathcal{R}|^2} \sum_{\mathbf{r}_i \in \mathcal{R}} \left(p^*(\mathbf{r}_i) - \frac{1}{|\mathcal{R}|} \sum_{\mathbf{r}_j \in \mathcal{R}} p^*(\mathbf{r}_j) \right) \cdot \left(-\sum_{\mathbf{a} \in \mathcal{A}} \log(\pi_{\theta(0)}(\mathbf{a} | \mathbf{r}_i, \mathbf{q})) \right). \end{aligned}$$

762 As we consider the case $|\mathcal{R}| \rightarrow +\infty$ and the success rate $p^*(\mathbf{r})$ are i.i.d. samples from $U[0, 1]$.
 763 By the Central limit theorem, for large $|\mathcal{R}|$, the sample mean $\frac{1}{|\mathcal{R}|} \sum_{\mathbf{r} \in \mathcal{R}} p^*(\mathbf{r})$ is concentrated
 764 around the its expectation 0.5, the deviation is of order is $\mathcal{O}(1/\sqrt{|\mathcal{R}|})$. We define this deviation as
 765 $\delta = \mathcal{O}(1/|\mathcal{R}|^{1/2}) > 0$ such that $\frac{1}{|\mathcal{R}|} \sum_{\mathbf{r} \in \mathcal{R}} p^*(\mathbf{r}) = 0.5 + \delta$.

766 According to the Gibbs conditioning principle ([12] Corollary 7.3.5 and Theorem 7.3.8), as $|\mathcal{R}| \rightarrow \infty$,
 767 the empirical distribution of the $p^*(\mathbf{r})$ conditioned on their mean $\frac{1}{|\mathcal{R}|} \sum_{\mathbf{r} \in \mathcal{R}} p^*(\mathbf{r}) = 0.5 + \delta$
 768 converges to the density: $f(x) = 1 + 12\delta(x - 0.5), x \in [0, 1]$. As $|\mathcal{R}| \rightarrow \infty$, we can approximate
 769 the summation over $\mathbf{r}$ with an integral over this limiting distribution $f(x)$. The derivative becomes:

$$\left. \frac{d}{dt} \text{ACC}_{\theta(t)} \right|_{t=0} = \int_0^1 - (1 + 12\delta(x - 0.5)) \cdot (x - 0.5 - \delta) \cdot \log(x(1 - x)) dx = \frac{2}{3}\delta > 0$$

770 Since $\delta > 0$, the derivative is positive. As $|\mathcal{R}| \rightarrow \infty$, this holds with probability 1, proving the
 771 first statement.

To prove the second statement, we first analyze the RLIF reward function. For an action space of size
 $|\mathcal{A}| = 2$, the reward for a path $\mathbf{r}$ is proportional to the negative log-likelihood of the policy:

$$R_{\text{RLIF}}(\mathbf{r}) - \frac{1}{|\mathcal{A}|} \sum_{\mathbf{a}_j \in \mathcal{A}} \log(\pi_{\theta}(\mathbf{a}_j | \mathbf{q}, \mathbf{r})) = -\frac{1}{2} \log(p^*(\mathbf{r})(1 - p^*(\mathbf{r})))$$

772 The function $g(p) = p(1 - p)$ is maximized at $p = 0.5$. Therefore, maximizing the reward $R_{\text{RLIF}}(\mathbf{r})$
 773 is equivalent to choosing the path whose success rate $p^*(\mathbf{r})$ is furthest from 0.5.

Let $p_{\min}^* = \min_{\mathbf{r} \in \mathcal{R}} p^*(\mathbf{r})$ and $p_{\max}^* = \max_{\mathbf{r} \in \mathcal{R}} p^*(\mathbf{r})$. The path with the highest reward will be the
 one with success rate $p_{\min}^*$ if it is further from 0.5 than $p_{\max}^*$ is. This condition is expressed as:

$$0.5 - p_{\min}^* > p_{\max}^* - 0.5 \iff p_{\min}^* + p_{\max}^* < 1$$

774 Thus, the event $\arg \max_{\mathbf{r}} R_{\text{RLIF}}(\mathbf{r}) = \arg \min_{\mathbf{r}} p^*(\mathbf{r})$ is equivalent to the event $p_{\min}^* + p_{\max}^* < 1$.

775 We now calculate the probability of this event. The success rates $p^*(\mathbf{r})$ are drawn from the limiting
 776 distribution $f(x) = 1 + 12\delta(x - 0.5)$. From the theory of order statistics, the scaled minimum and
 777 maximum of $n = |\mathcal{R}|$ samples converge in distribution to independent exponential random variables.
 778 Let $Y_n = n \cdot p_{\min}^*$ and $Z_n = n \cdot (1 - p_{\max}^*)$. As $n \rightarrow \infty$:

- 779 • Y_n converges to $Y \sim \text{Exp}(\lambda_Y)$, where the rate is the density at the lower bound: $\lambda_Y = f(0) =$
 780 $1 - 6\delta$.
- 781 • Z_n converges to $Z \sim \text{Exp}(\lambda_Z)$, where the rate is the density at the upper bound: $\lambda_Z = f(1) =$
 782 $1 + 6\delta$.

The condition $p_{\min}^* + p_{\max}^* < 1$ can be rewritten in terms of these asymptotic variables:

$$\frac{Y_n}{n} + \left(1 - \frac{Z_n}{n}\right) < 1 \implies Y_n < Z_n$$

The probability of this event is $P(Y < Z)$. For two independent exponential variables, this is given by:

$$P(Y < Z) = \frac{\lambda_Y}{\lambda_Y + \lambda_Z} = \frac{1 - 6\delta}{(1 - 6\delta) + (1 + 6\delta)} = \frac{1 - 6\delta}{2} = \frac{1}{2} - 3\delta$$

Since $\delta = O(1/|\mathcal{R}|^{1/2})$, in the limit as $|\mathcal{R}| \rightarrow \infty$, $\delta \rightarrow 0$. Therefore, the limiting probability is:

$$\lim_{|\mathcal{R}| \rightarrow \infty} P(\arg \max_{\mathbf{r}} R_{\text{RLIF}}(\mathbf{r}) = \arg \min_{\mathbf{r}} p^*(\mathbf{r})) = \lim_{\delta \rightarrow 0} \left(\frac{1}{2} - 3\delta \right) = \frac{1}{2}$$

783 This concludes the proof of the second statement.

784

□

785 D Full Prompts

786 D.1 Full Prompt for Reasoning Patterns Extraction

Full Prompt for Reasoning Patterns Extraction

You will analyze multiple AI reasoning processes, showing how different models solve problems.

Analysis Steps:

For each reasoning process, identify:

Key words and recurring phrases

Logical structure of the argument

Problem-solving techniques used

Step-by-step progression

Group similar reasoning processes into exactly 5 patterns based on shared:

Vocabulary patterns (common terms and phrases)

Logical frameworks (how arguments are structured)

Solution approaches (methods used to reach conclusions)

IMPORTANT! You need to categorize general thinking patterns for problem solving, rather than categorizing types of problems themselves. In other words, each type of thinking pattern you give can be applied to any problem, rather than being task-specific. What you need to classify is the way of thinking, not the way of solving problems.

For example, you can classify them into:

1. derivation based on existing knowledge

2. derivation combined with verification

3. independent calculation, suitable for learners starting from the basics,

4. detailed manual derivation, suitable for learners who need to deeply understand each step

5. etc

Or you can classify them into:

1. knowledge call type

2. derivation and verification type

3. etc.

Output Format:

Pattern 1:

Name and description

Defining characteristics

Example processes from input

Common elements

Pattern 2:

Name and description

Defining characteristics

Example processes from input

Common elements

...

There are reasoning processes:

787

{extracted_responses}

Group similar reasoning processes into exactly 5 patterns. Please provide a classification of reasoning patterns, with each group defined by their common characteristics and examples from these processes.

788

789 D.2 Full Prompt for Responses Classification

Full Prompt for Responses Classification

Please classify each reasoning process into one of the following five patterns. Output the classifications in JSON format.

{patterns_categories}

Here are the reasoning processes to classify:

{extracted_responses}

Please output your classification in this JSON format:

790

791 D.3 Prompt Examples for Evaluation Tasks

Prompt Example for Easy Countdown

<lim_start>system

You are a helpful assistant. You first thinks about the reasoning process in the mind and then provides the user with the answer.<lim_end>

<lim_start>

user Using the numbers [1,2,3,7], create an equation that equals 5. You can only use basic arithmetic addition (+) and subtraction (-), and each number from the list can be used at most once. Show your work in <think> </think> tags. And return the final answer in <answer> </answer> tags, for example, to get 5 from [1,2,3,7], we have <answer> 2 + 3 = 5 </answer>; to get 13 from [1,2,3,7], we have <answer> 1 + 2 + 3 + 7 = 13 </answer>.<lim_end>

<lim_start>assistant

Let me solve this step by step.

<think>

792

Prompt Example for Long Arithmetic

<lim_start>system

You are a helpful assistant. You first thinks about the reasoning process in the mind and then provides the user with the answer.<lim_end>

<lim_start>user

Considering the experssion (((((((((16 + 47) + 92) + 84) + 5) + 49) + 82) - 33) - 99) - 69), you need to start from the innermost bracket and work your way out. You need to give the specific calculation process. You have to do a certain amount of calculations on each line, and print each calculation in the format of a + b = c or a - b = c, for example "1 + 2 = 3".<lim_end>

<lim_start>assistant

Let me solve this step by step.

<think>

793

Prompt Example for Number Theory

<lim_start>system

You are a helpful assistant.<lim_end>

<lim_start>user

You have four bags of copper coins. Each bag has the same number of copper coins. One day, you find a bag of 23 coins. You decide to redistribute the number of coins you have

794

so that all five bags you hold have the same number of coins. You successfully manage to redistribute all the coins, and you also note that you have more than 120 coins. What is the smallest number of coins you could have had before finding the bag of 23 coins?
Please reason step by step, and put your final answer within $\boxed{}$.<lim_end>
<lim_start>assistant

795

Prompt Example for Geometry

<lim_start>system
You are a helpful assistant.<lim_end>
<lim_start>user
Two sides of scalene $\triangle ABC$ measure 4 centimeters and 7 centimeters. How many different whole centimeter lengths are possible for the third side?
Please reason step by step, and put your final answer within $\boxed{}$.<lim_end>
<lim_start>assistant

796

Prompt Example for Algebra

<lim_start>system
You are a helpful assistant.<lim_end>
<lim_start>user
Six aluminum cans can be recycled to make a new can. How many new cans can eventually be made from 200 aluminum cans? (Remember that the first new cans that are made can then be recycled into even newer cans!)
Please reason step by step, and put your final answer within $\boxed{}$.<lim_end>
<lim_start>assistant

797

Prompt Example for Calculus

<lim_start>system
You are a helpful assistant.<lim_end>
<lim_start>user
For how many values of x in $[0, 2\pi]$ is $\cos^{-1}(\cos 4x) = \sin^{-1}(\sin x)$?
Please reason step by step, and put your final answer within $\boxed{}$.<lim_end>
<lim_start>assistant

798

Prompt Example for Counting and Probability

<lim_start>system
You are a helpful assistant.<lim_end>
<lim_start>user
7 white balls and k black balls are placed into a bin. Two balls are drawn at random. The probability that one ball is white and the other is black is $\frac{35}{66}$. Find the smallest possible value of k .
Please reason step by step, and put your final answer within $\boxed{}$.<lim_end>
<lim_start>assistant

799

Prompt Example for AMC23-Q19

<lim_start>system
You are a helpful assistant.<lim_end>
<lim_start>user
How many digits are in the base-ten representation of $8^5 \cdot 5^{10} \cdot 15^5$?
Please reason step by step, and put your final answer within $\boxed{}$.<lim_end>
<lim_start>assistant

800

Prompt Example for AIME24-Q1

<lim_start>system
You are a helpful assistant.<lim_end>
<lim_start>user
Every morning Aya goes for a s -kilometer-long walk and stops at a coffee shop afterwards. When she walks at a constant speed of s kilometers per hour, the walk takes her 4 hours, including t minutes spent in the coffee shop. When she walks $s+2$ kilometers per hour, the walk takes her 2 hours and 24 minutes, including t minutes spent in the coffee shop. Suppose Aya walks at $s+\frac{1}{2}$ kilometers per hour. Find the number of minutes the walk takes her, including the t minutes spent in the coffee shop.
Please reason step by step, and put your final answer within $\boxed{}$.<lim_end>
<lim_start>assistant

801

Prompt Example for OlympiadBench-Q1631

<lim_start>system
You are a helpful assistant.<lim_end>
<lim_start>user
For a positive integer a , define a sequence of integers $x_1, x_2, \dots$ by letting $x_1 = a$ and $x_{n+1} = 2x_n + 1$ for $n \geq 1$. Let $y_n = 2^{x_n} - 1$. Determine the largest possible k such that, for some positive integer a , the numbers $y_1, \dots, y_k$ are all prime.
Please reason step by step, and put your final answer within $\boxed{}$.<lim_end>
<lim_start>assistant

802